

BFS Traversal

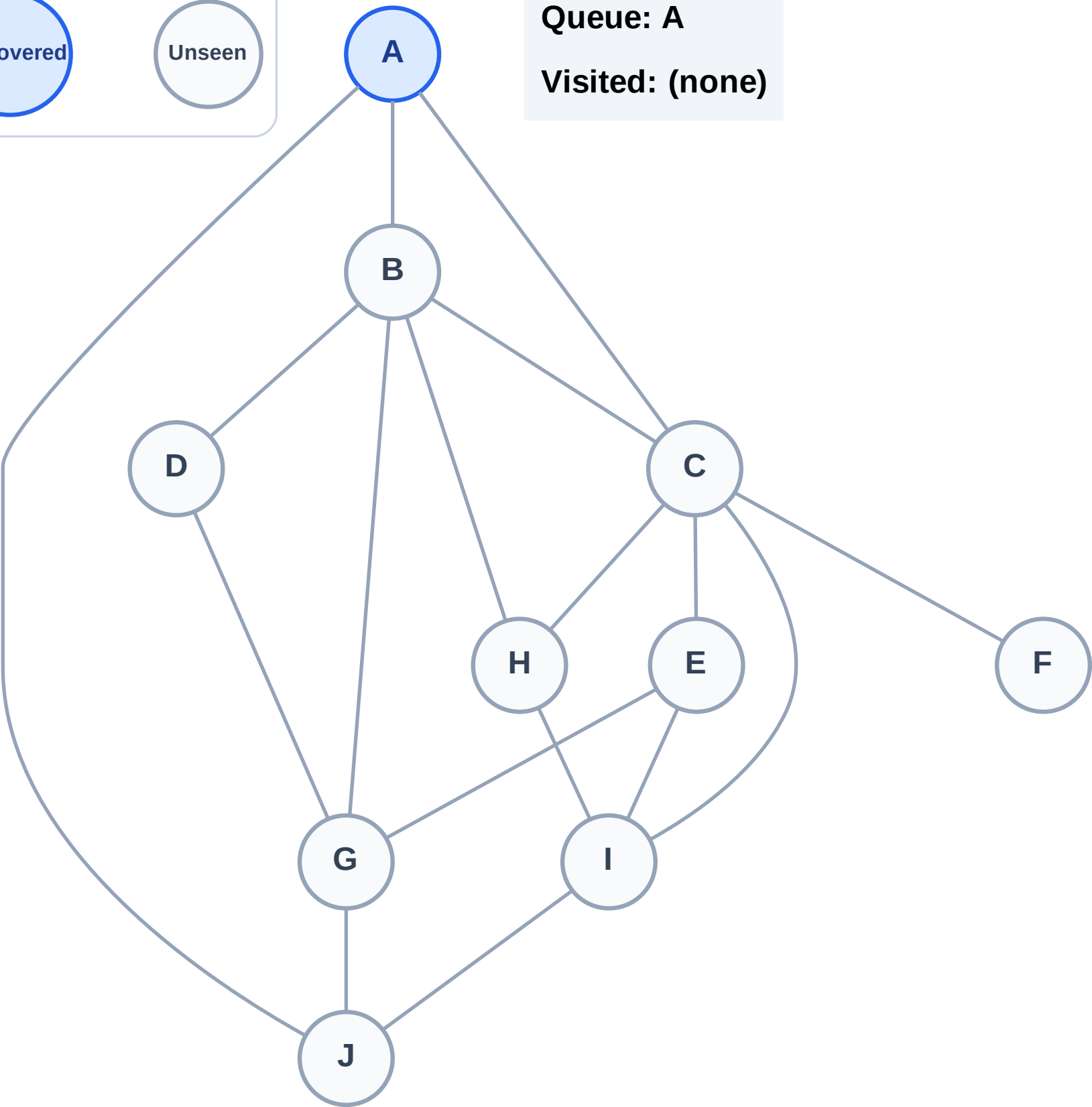
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

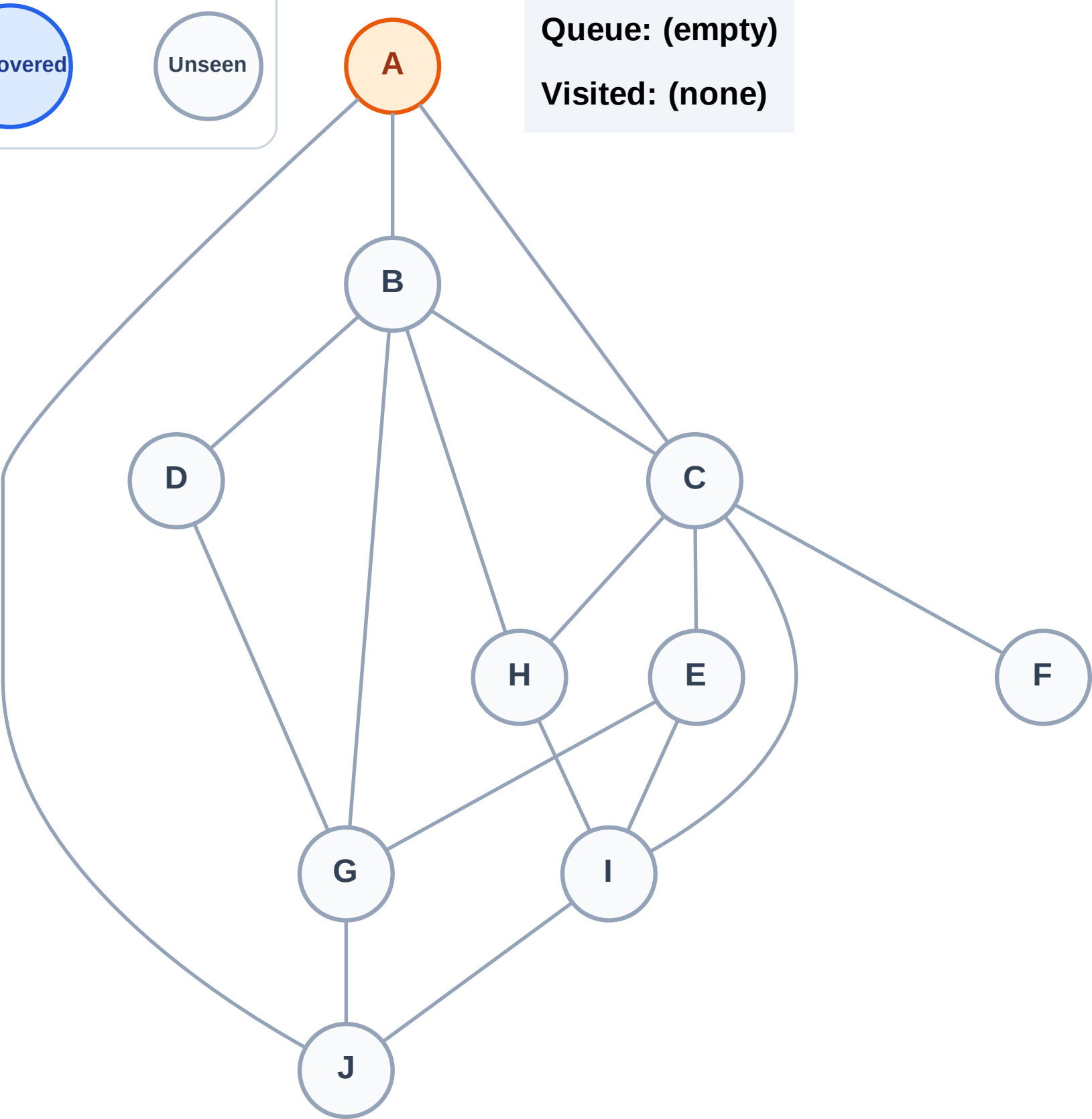
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

Step 3: Discover B, C, J from A

Legend

Complete

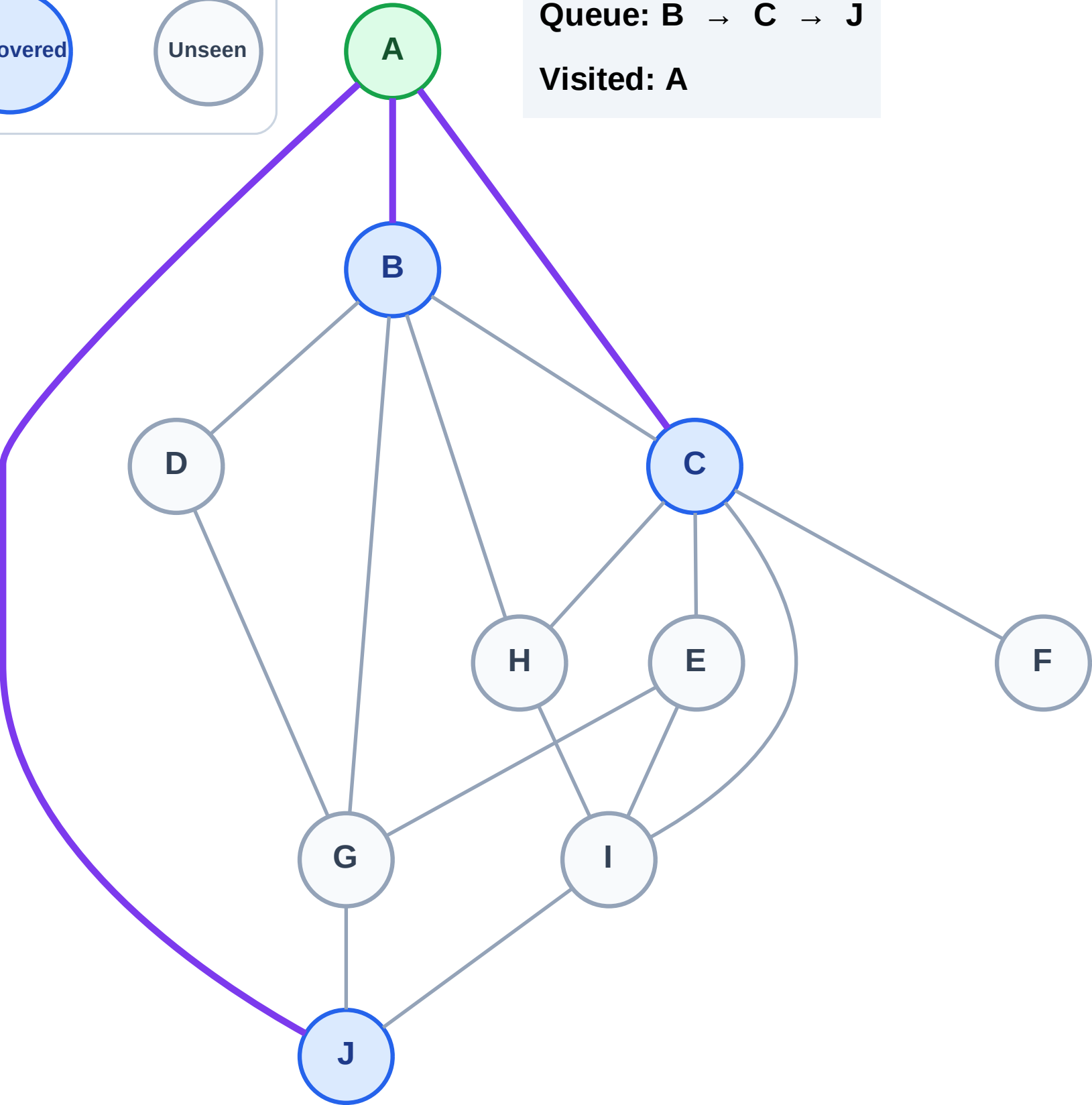
Processing

Discovered

Unseen

Queue: B → C → J

Visited: A



BFS Traversal

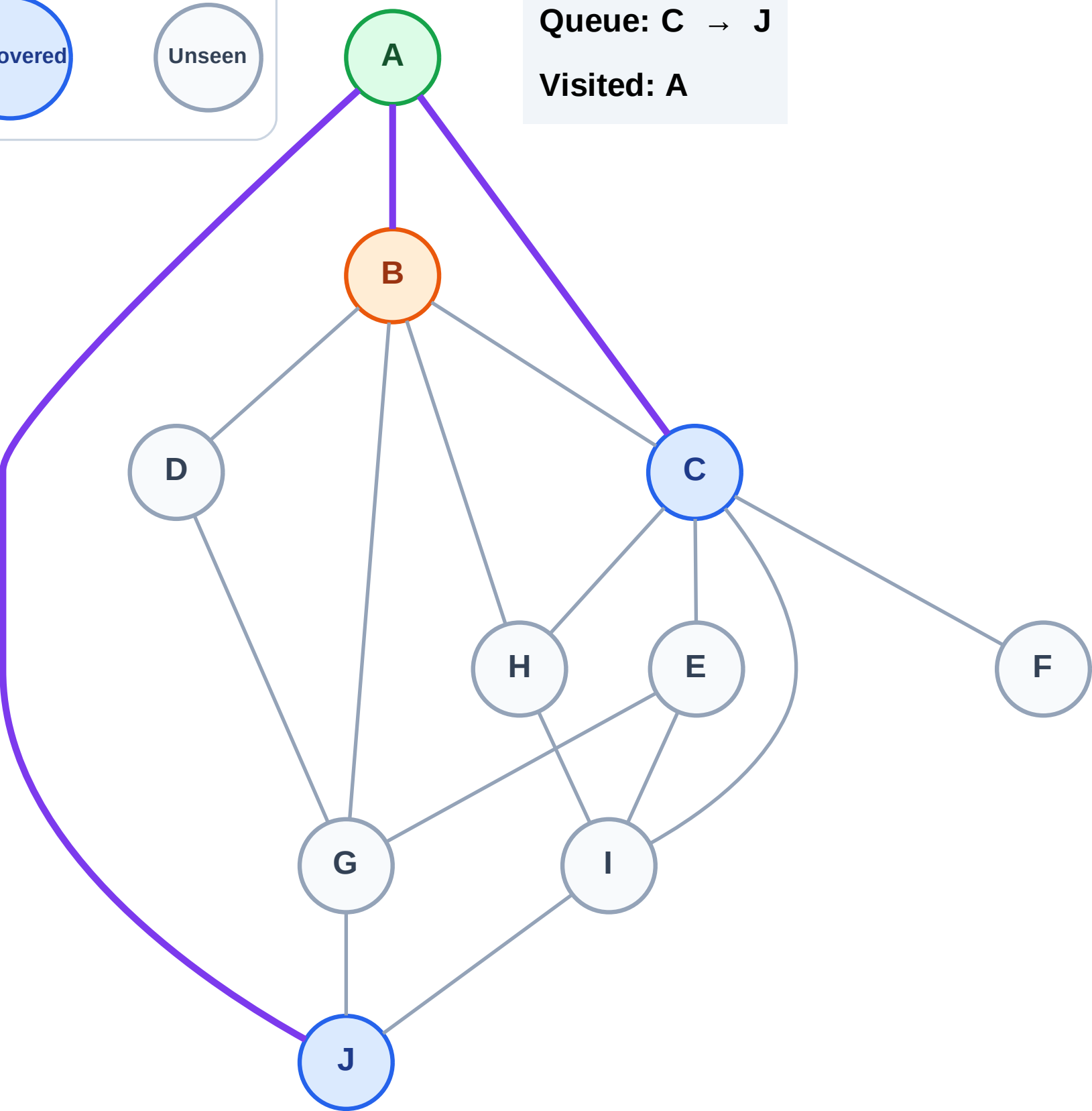
Step 4: Process node B

Legend



Queue: C → J

Visited: A



BFS Traversal

Step 5: Discover D, G, H from B

Legend

Complete

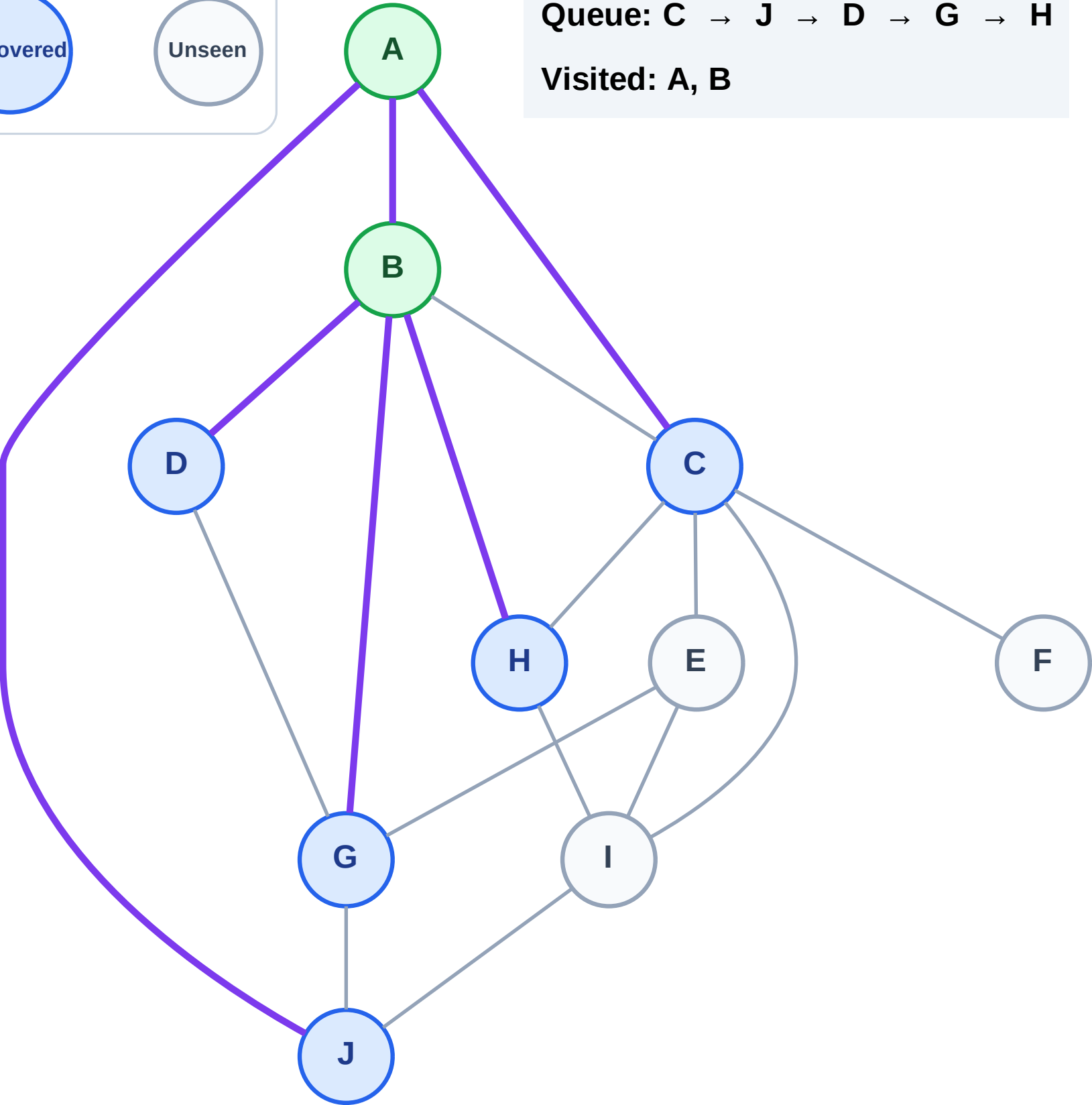
Processing

Discovered

Unseen

Queue: C → J → D → G → H

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

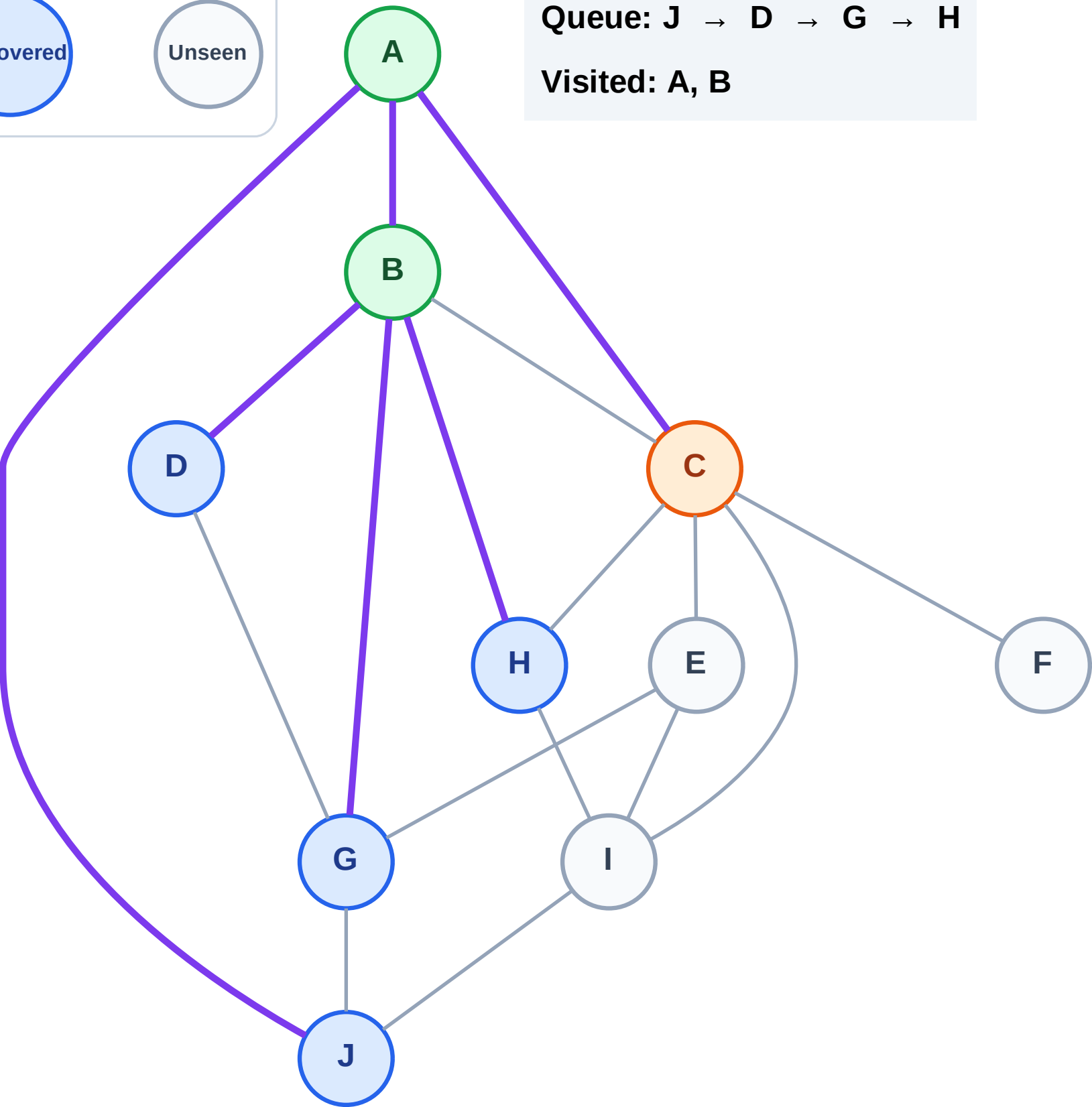
Processing

Discovered

Unseen

Queue: J → D → G → H

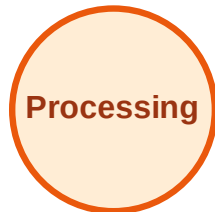
Visited: A, B



BFS Traversal

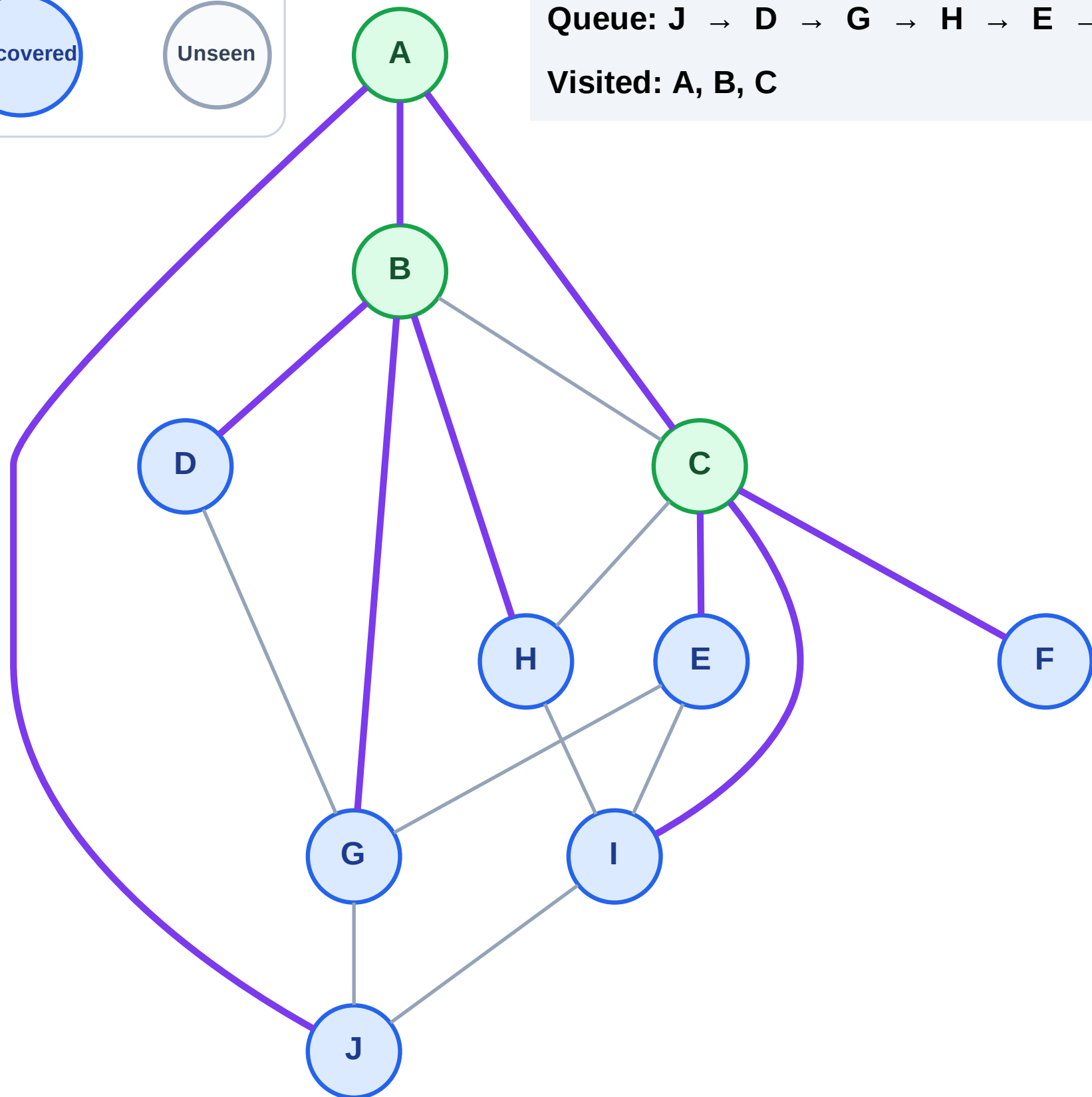
Step 7: Discover E, F, I from C

Legend



Queue: J → D → G → H → E → F → I

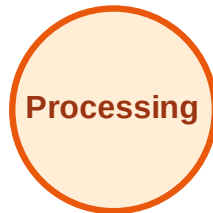
Visited: A, B, C



BFS Traversal

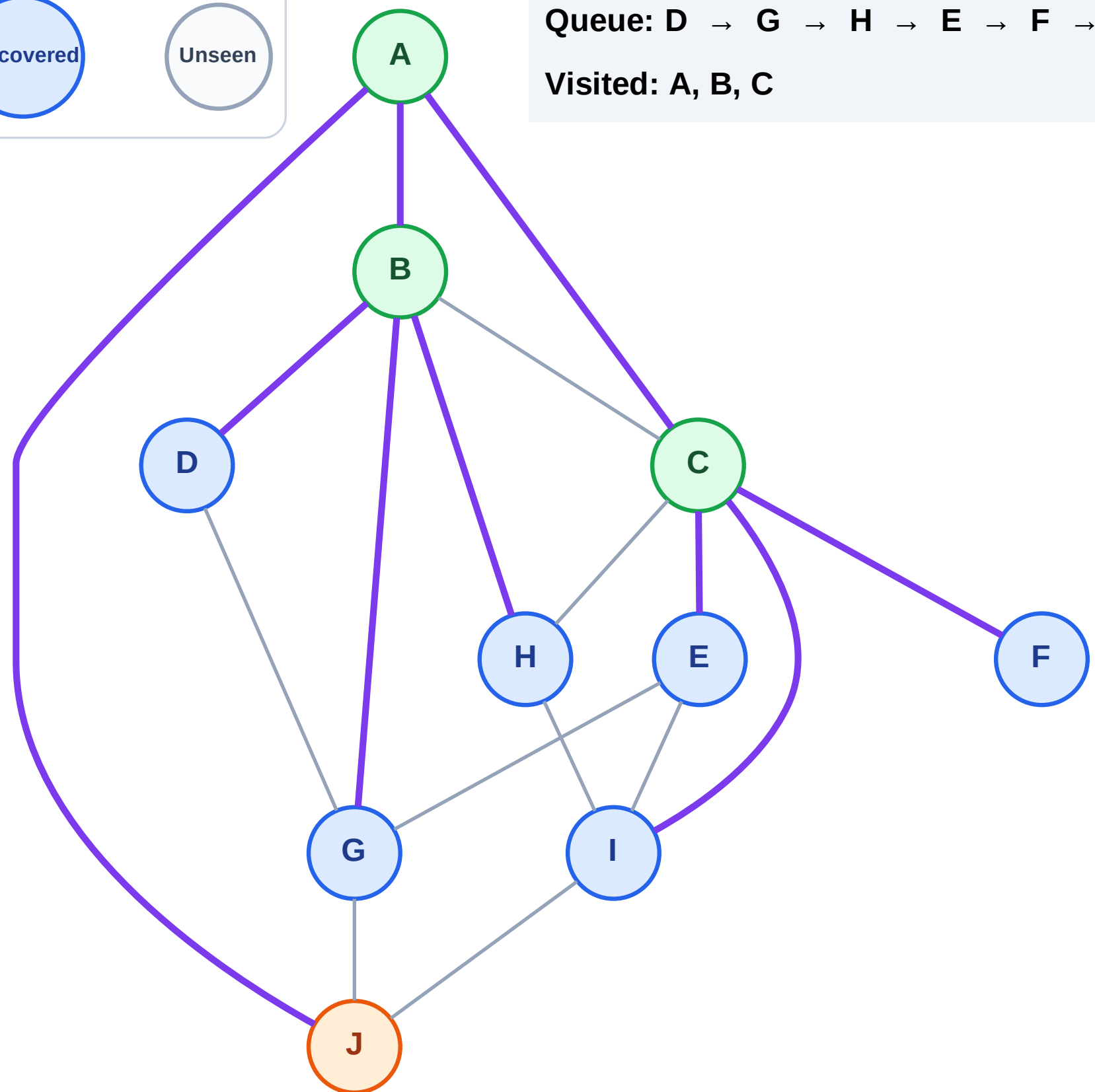
Step 8: Process node J

Legend



Queue: D → G → H → E → F → I

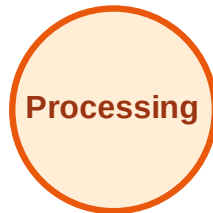
Visited: A, B, C



BFS Traversal

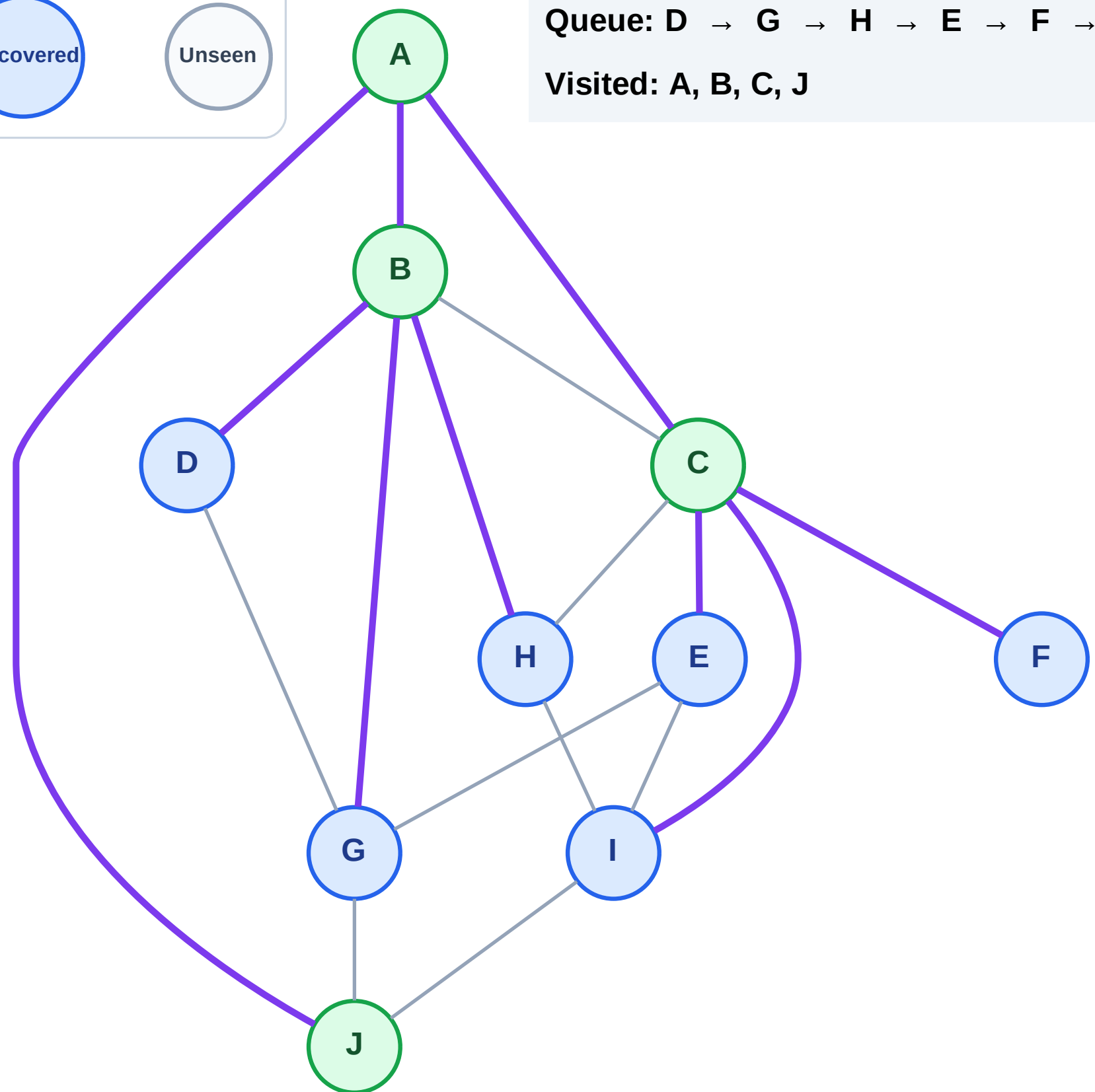
Step 9: No new neighbors from J

Legend



Queue: D → G → H → E → F → I

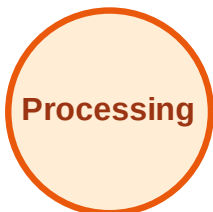
Visited: A, B, C, J



BFS Traversal

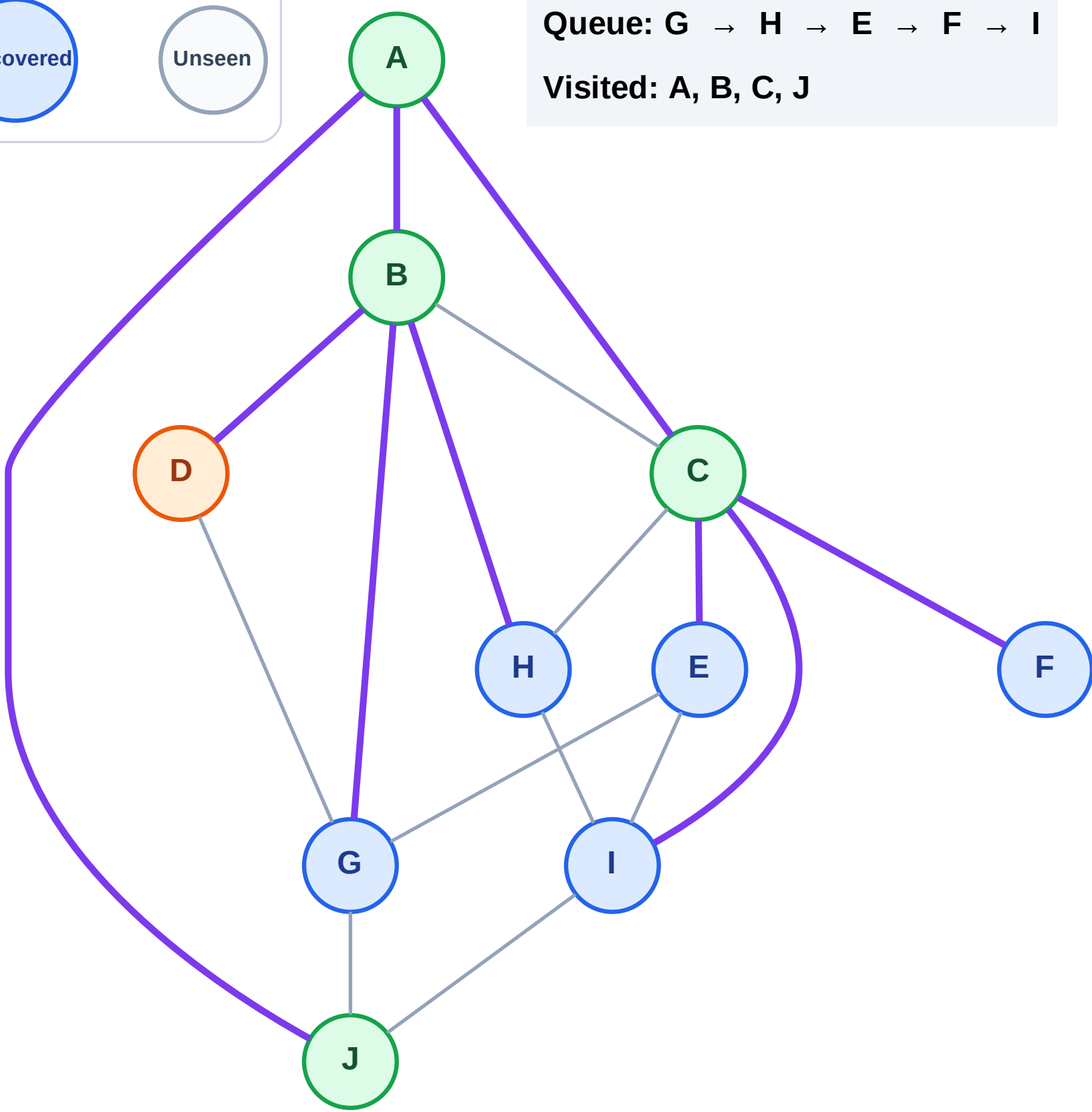
Step 10: Process node D

Legend



Queue: G → H → E → F → I

Visited: A, B, C, J



BFS Traversal

Step 11: No new neighbors from D

Legend

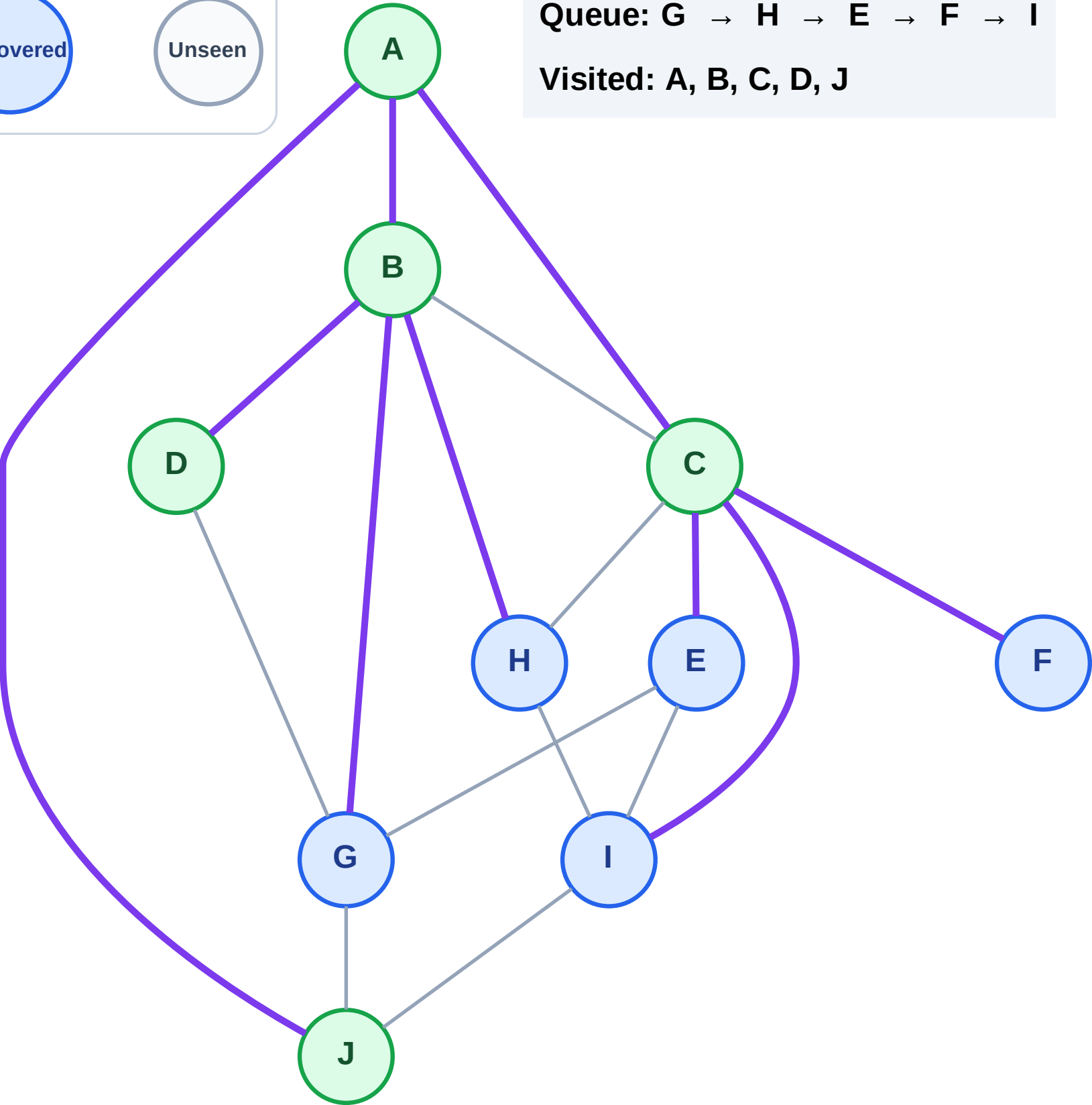
Complete

Processing

Discovered

Unseen

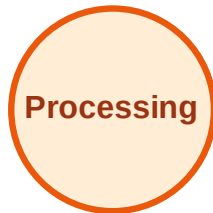
Queue: G → H → E → F → I
Visited: A, B, C, D, J



BFS Traversal

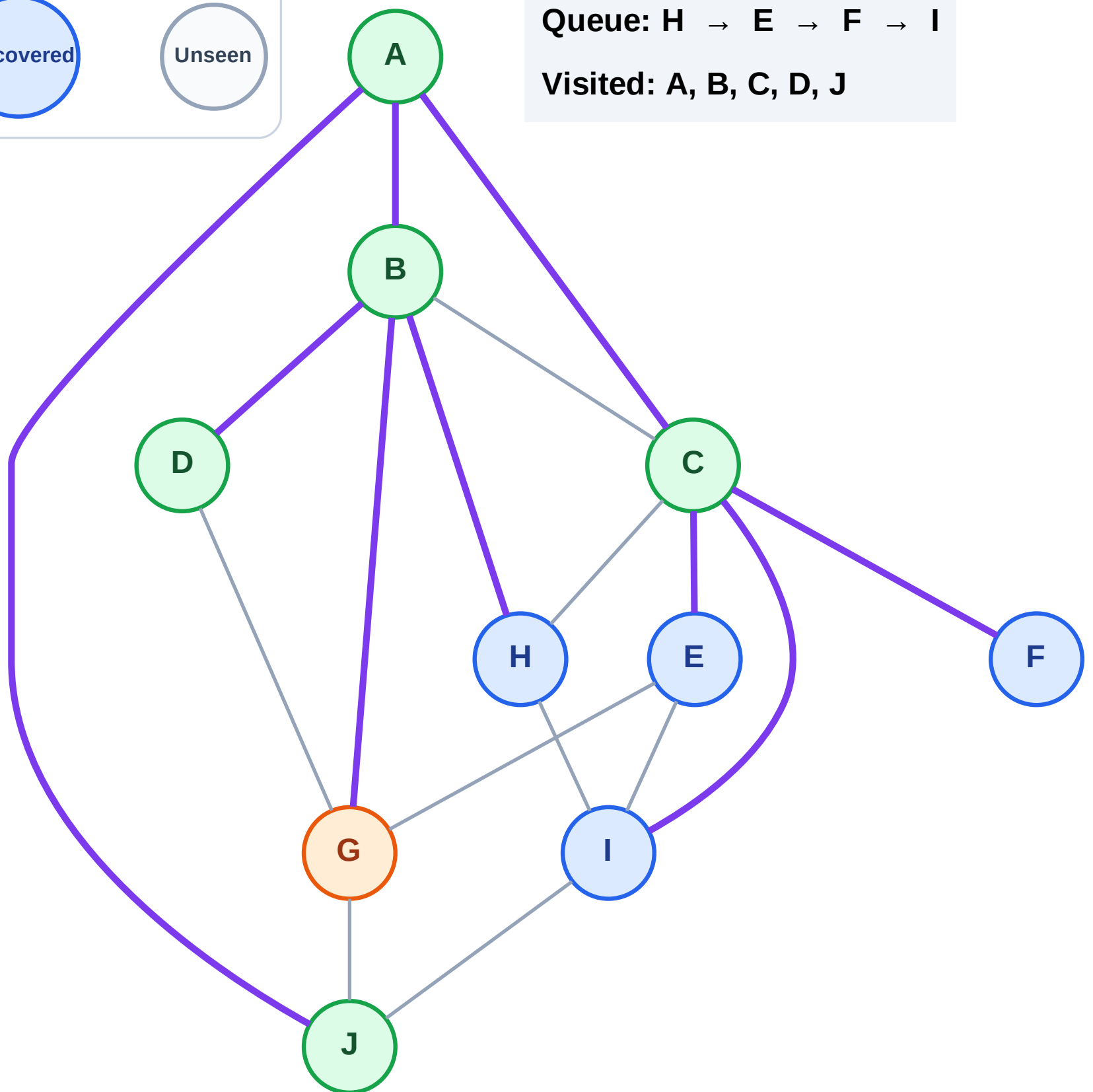
Step 12: Process node G

Legend



Queue: H → E → F → I

Visited: A, B, C, D, J



BFS Traversal

Step 13: No new neighbors from G

Legend

Complete

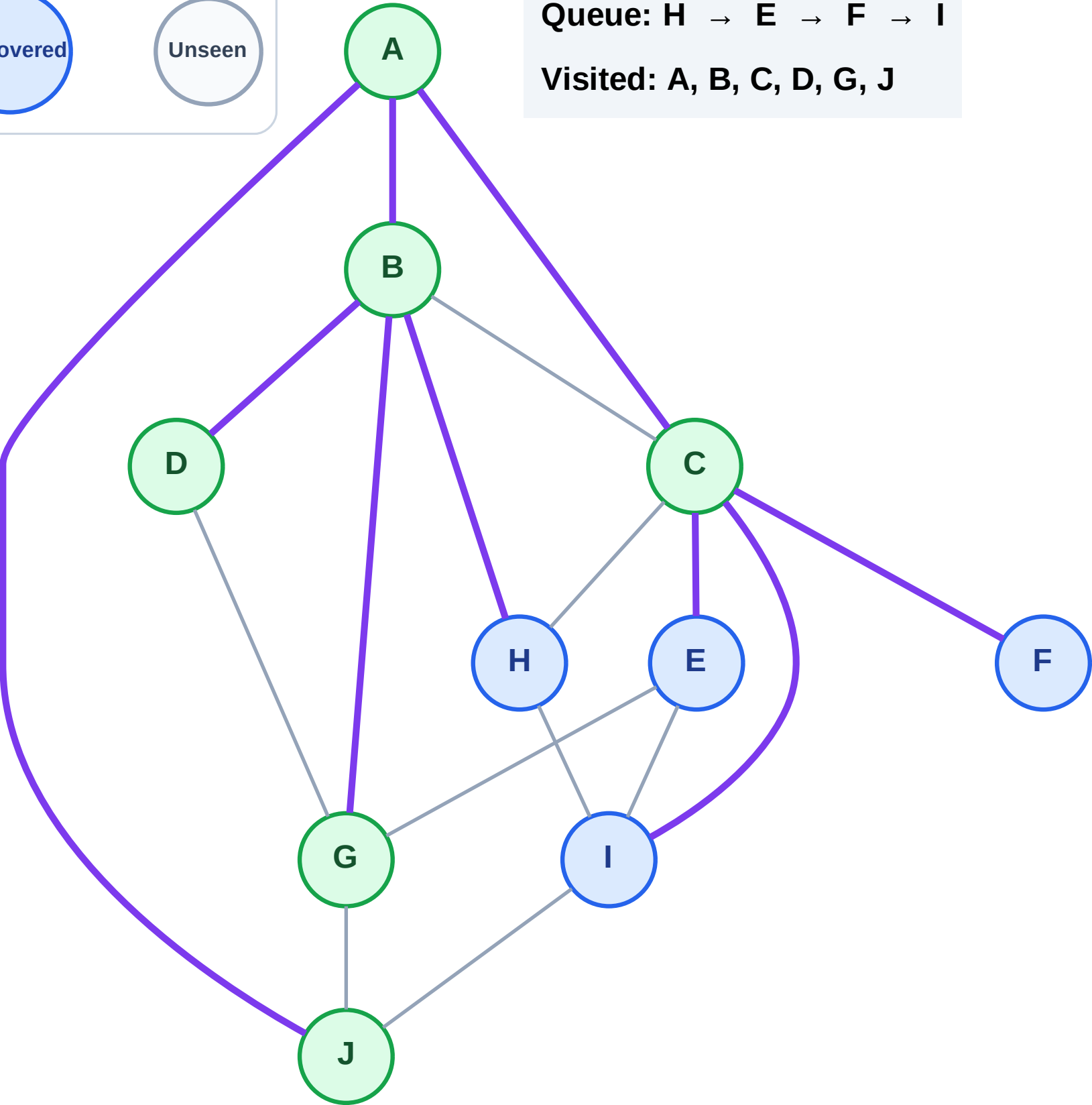
Processing

Discovered

Unseen

Queue: H → E → F → I

Visited: A, B, C, D, G, J



BFS Traversal

Step 14: Process node H

Legend

Complete

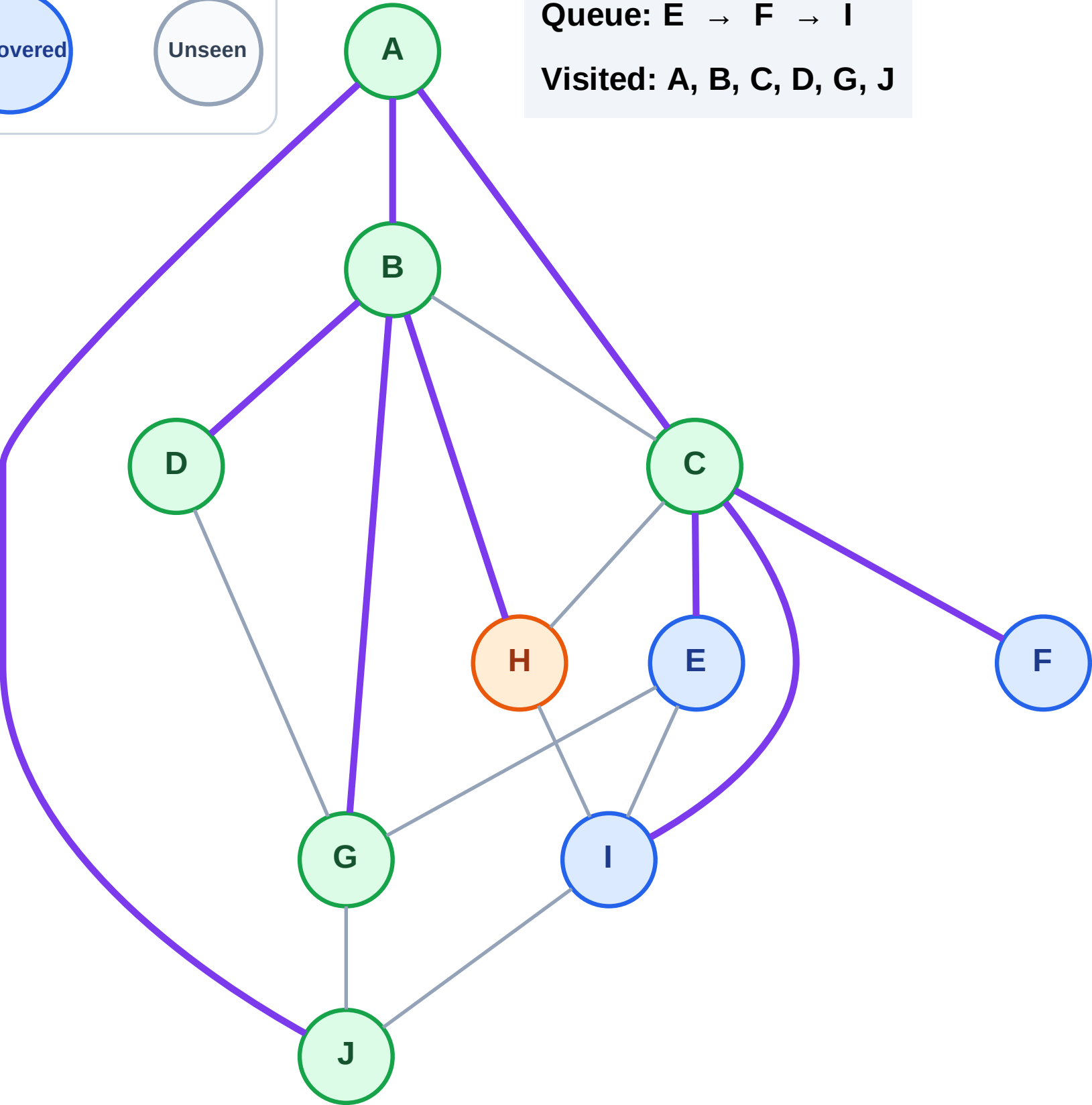
Processing

Discovered

Unseen

Queue: E → F → I

Visited: A, B, C, D, G, J



BFS Traversal

Step 15: No new neighbors from H

Legend

Complete

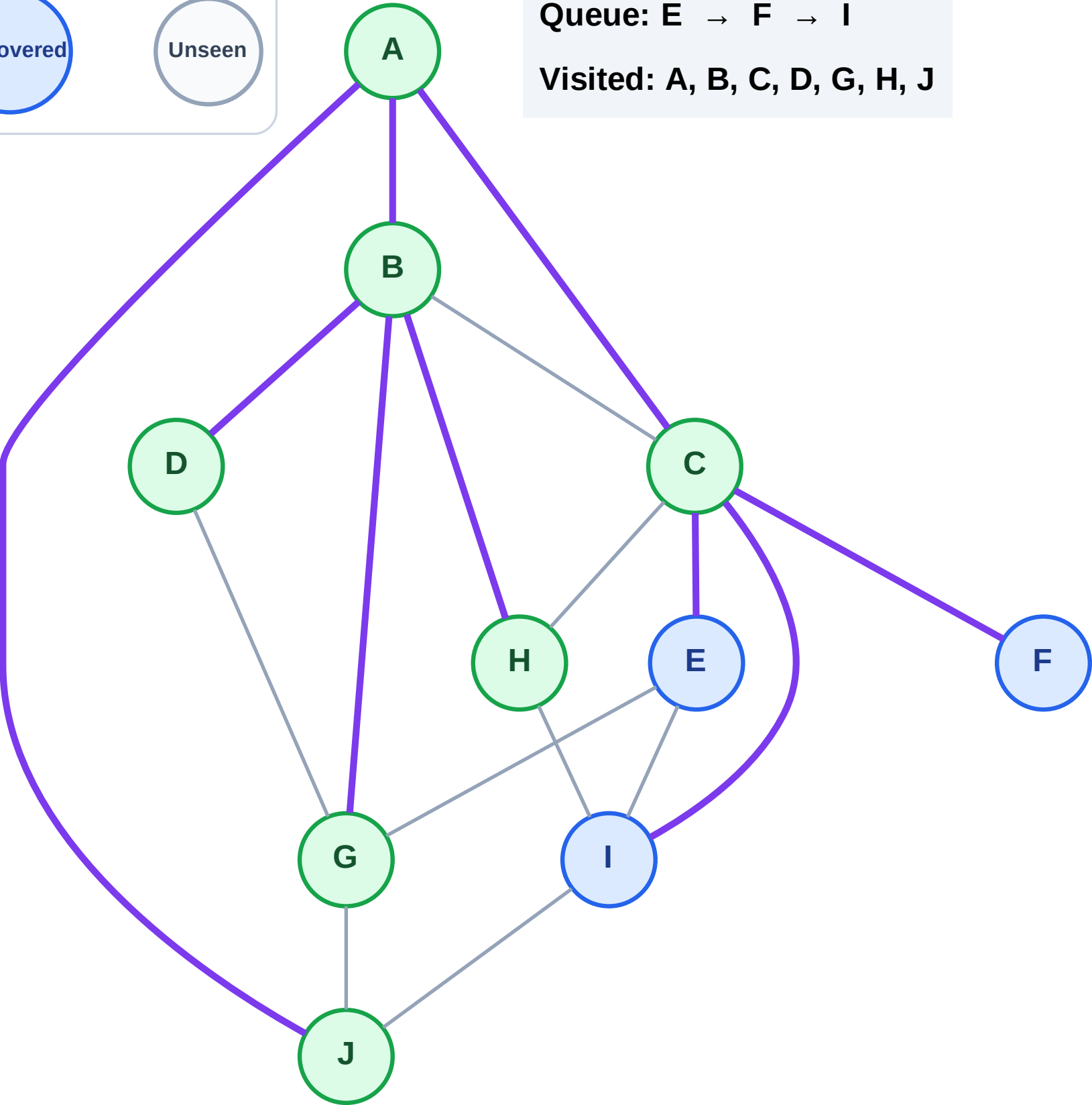
Processing

Discovered

Unseen

Queue: E → F → I

Visited: A, B, C, D, G, H, J



BFS Traversal

Step 16: Process node E

Legend

Complete

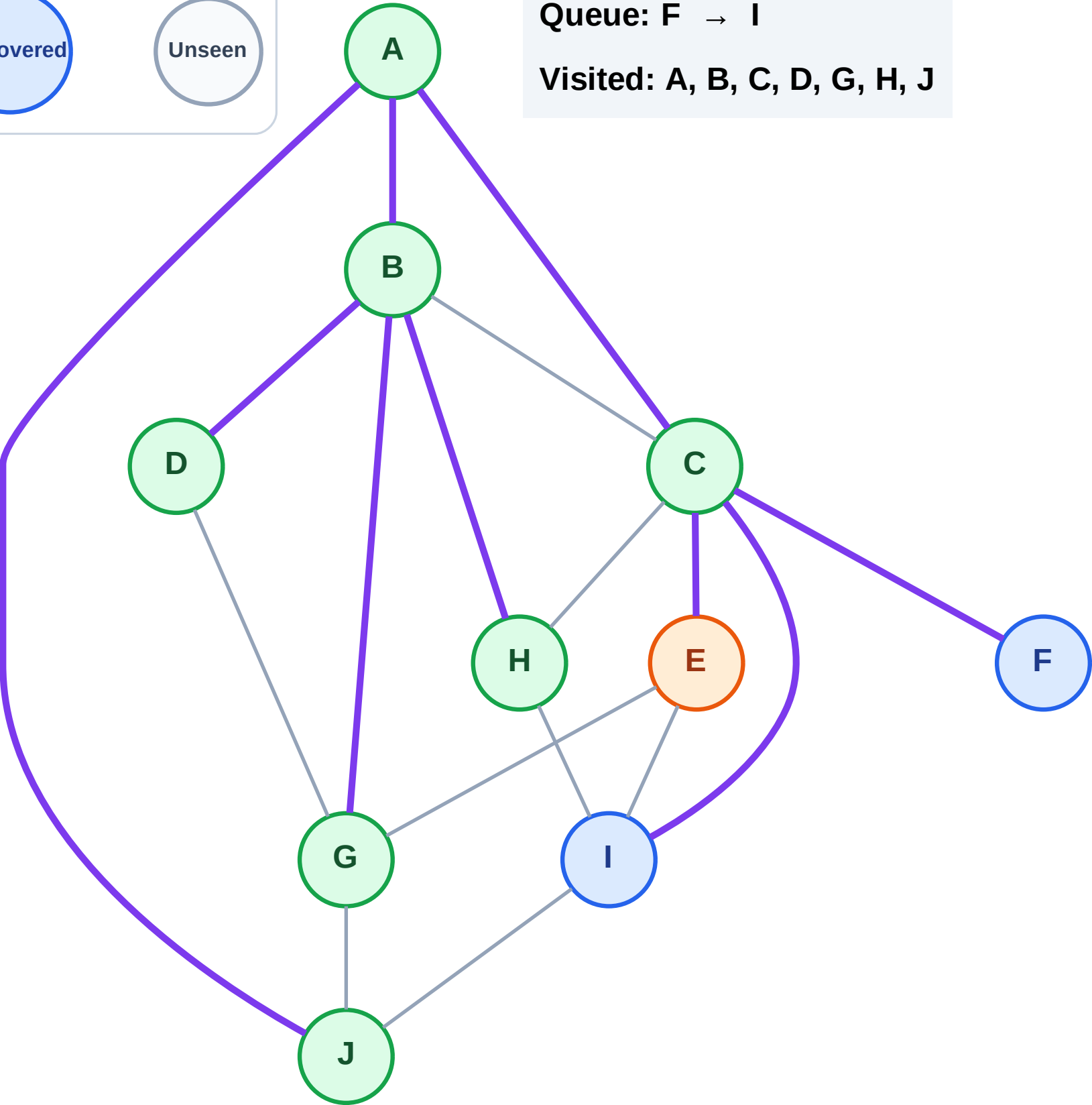
Processing

Discovered

Unseen

Queue: F → I

Visited: A, B, C, D, G, H, J



BFS Traversal

Step 17: No new neighbors from E

Legend

Complete

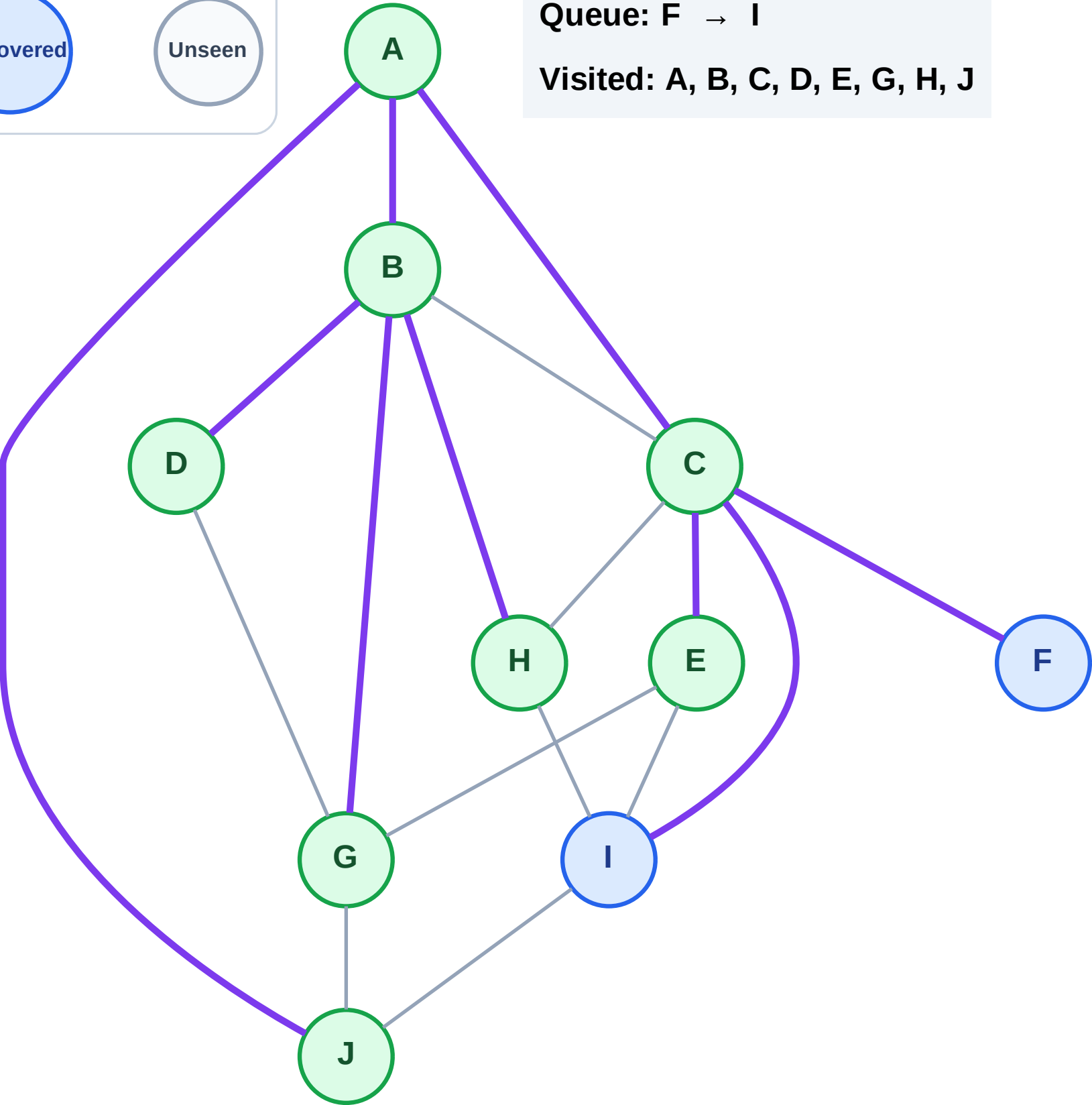
Processing

Discovered

Unseen

Queue: F → I

Visited: A, B, C, D, E, G, H, J



BFS Traversal

Step 18: Process node F

Legend

Complete

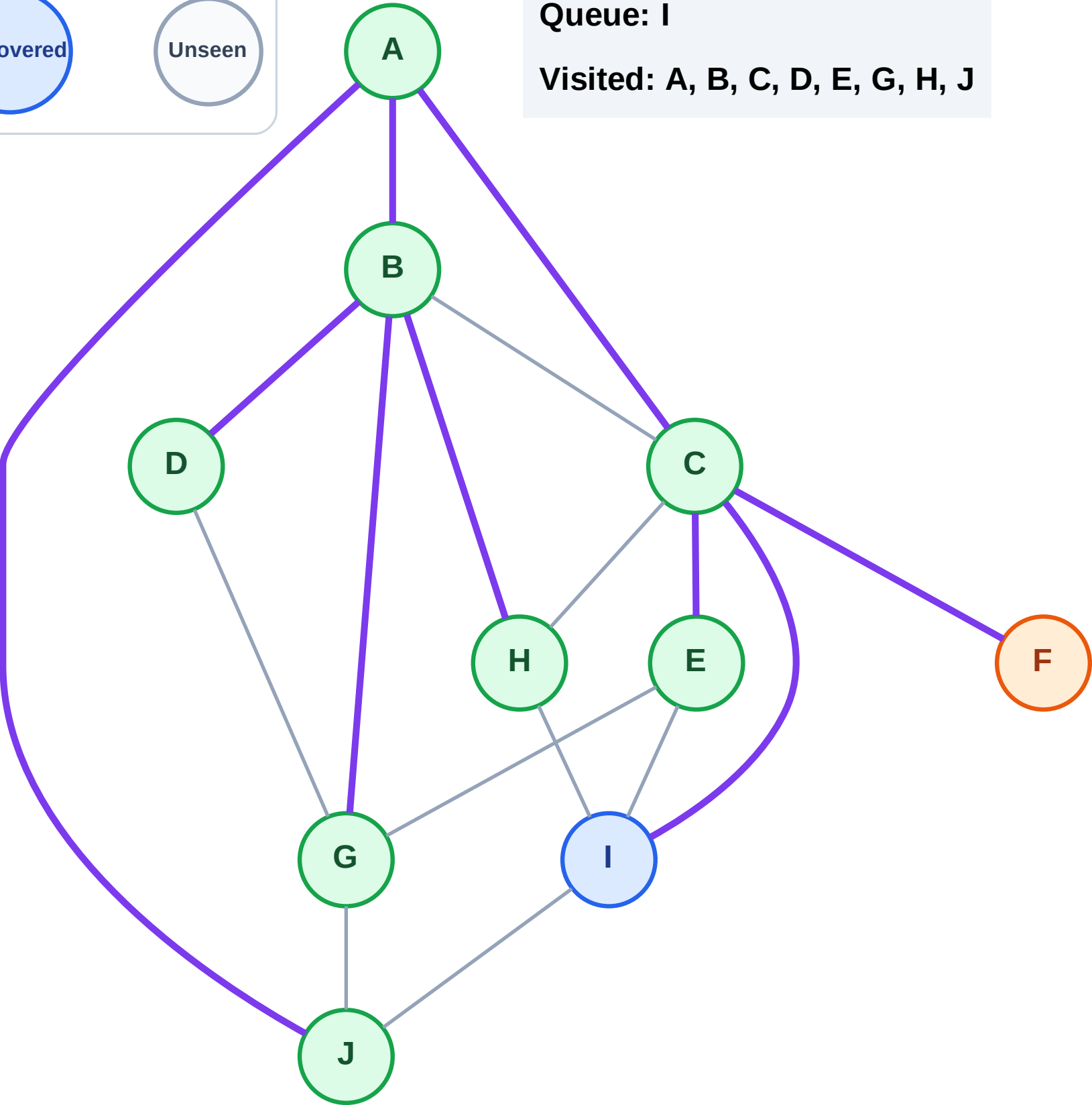
Processing

Discovered

Unseen

Queue: I

Visited: A, B, C, D, E, G, H, J



BFS Traversal

Step 19: No new neighbors from F

Legend

Complete

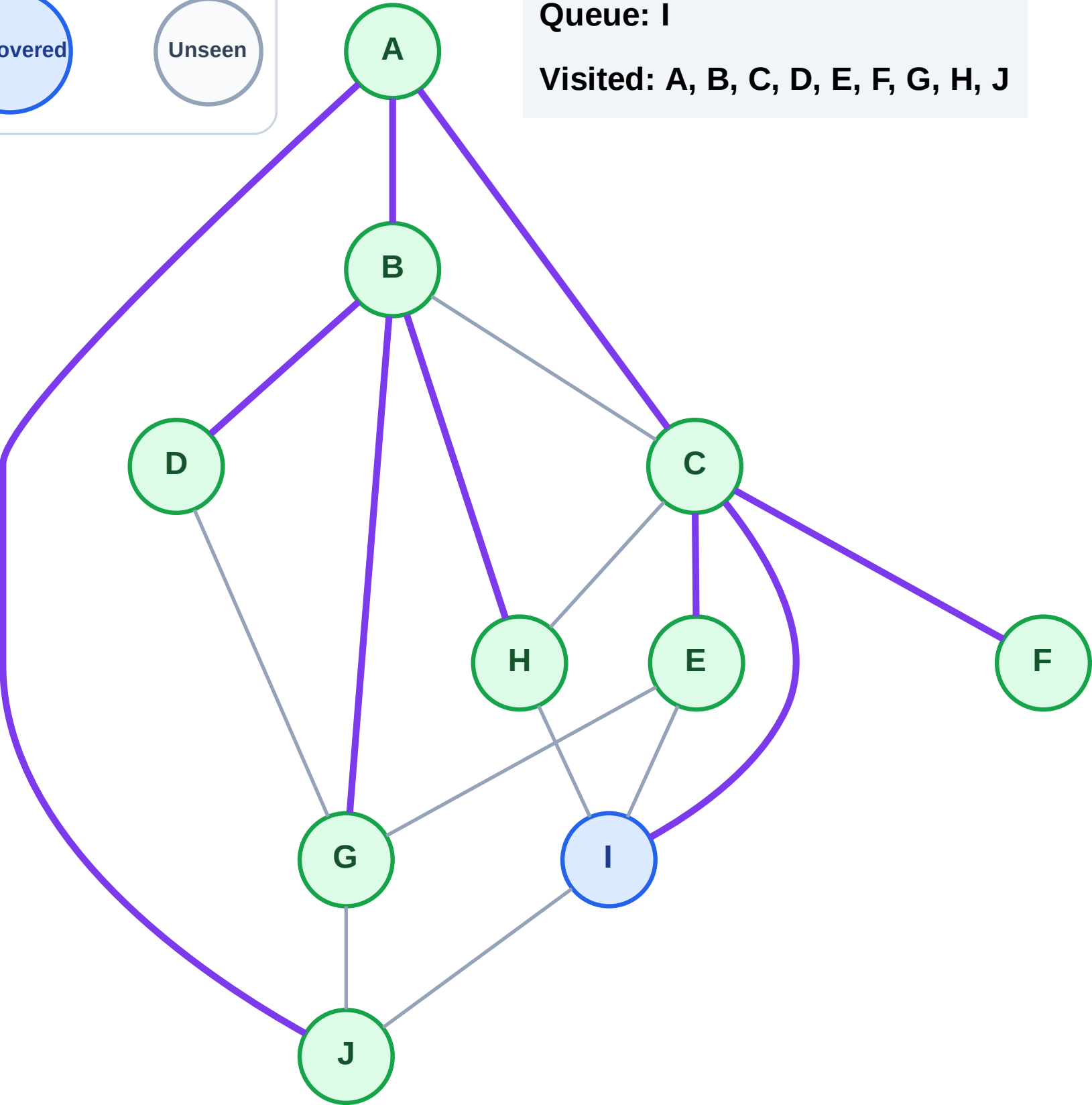
Processing

Discovered

Unseen

Queue: I

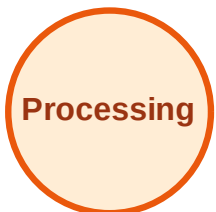
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

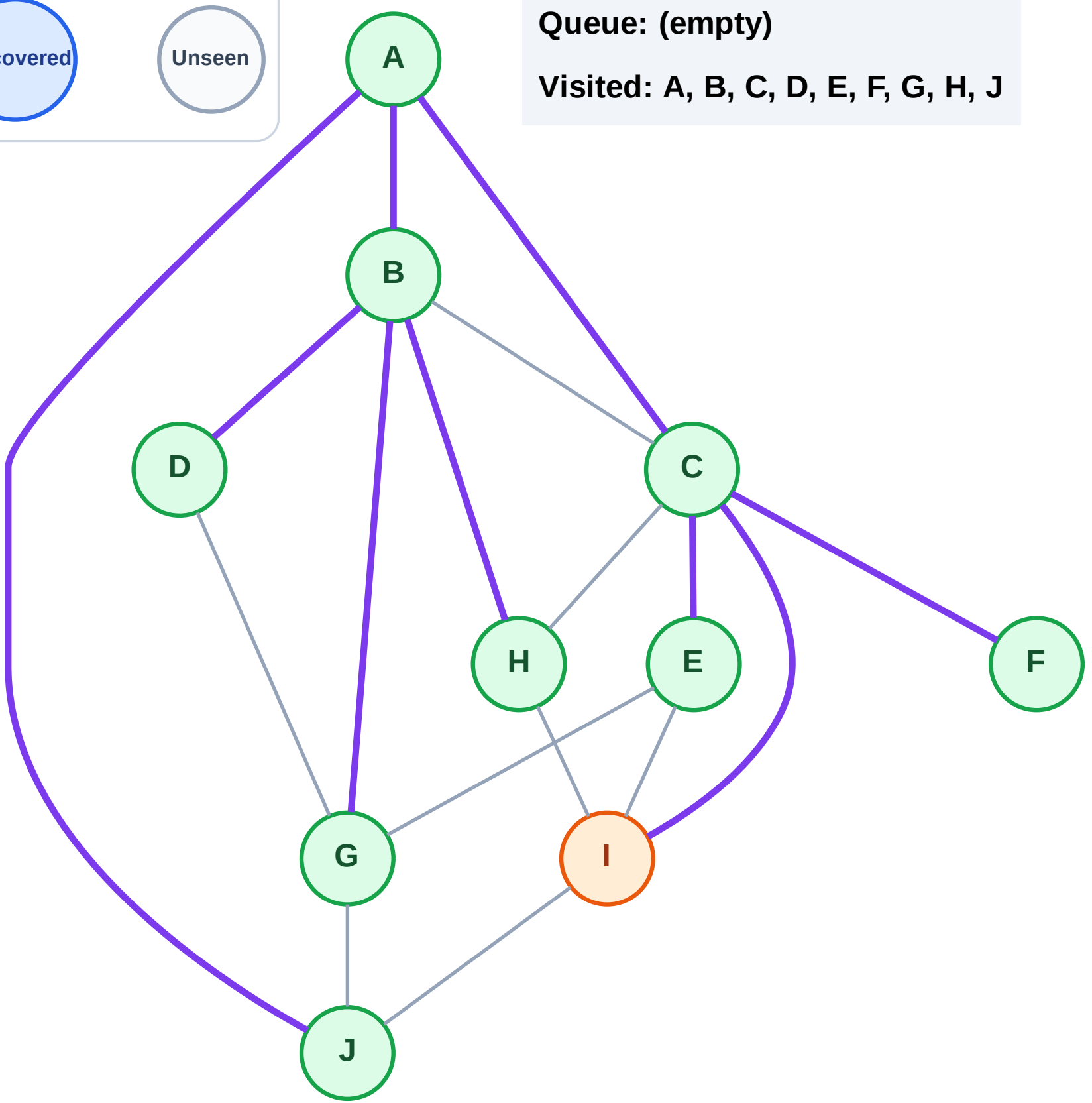
Step 20: Process node I

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

Step 21: No new neighbors from I

Legend

Complete

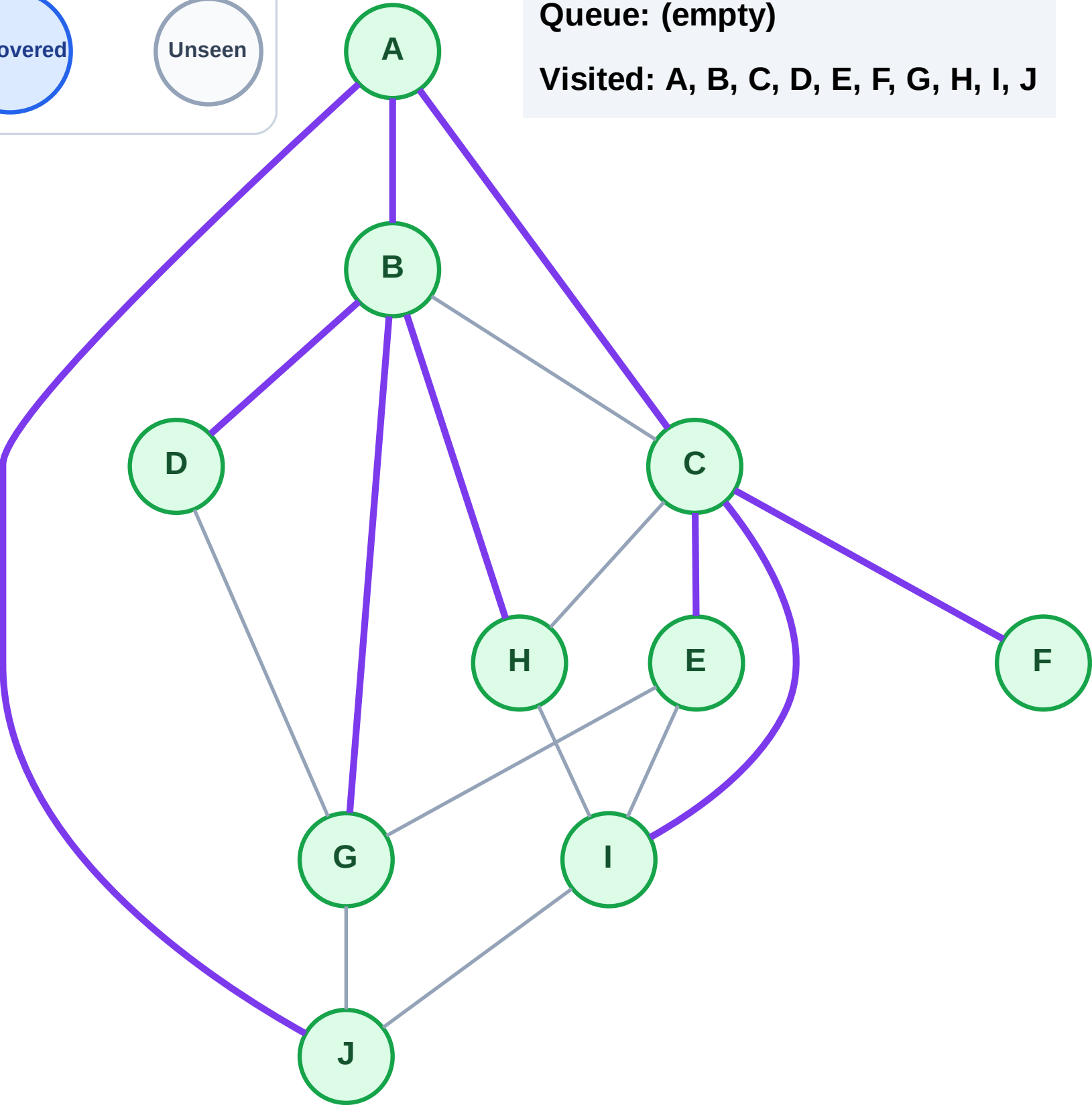
Processing

Discovered

Unseen

Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

Step 1: Start at A

Legend

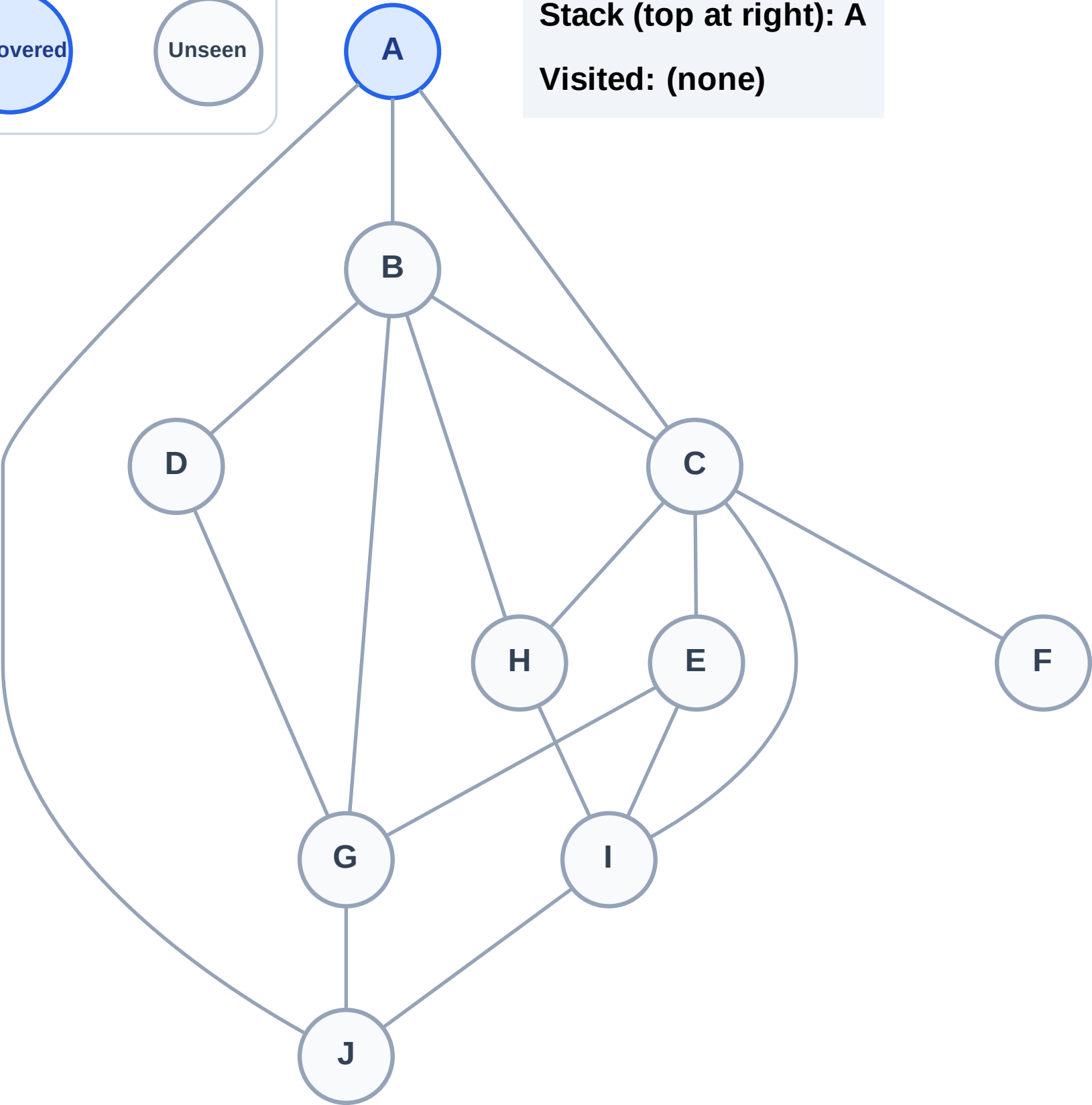
Complete

Processing

Discovered

Unseen

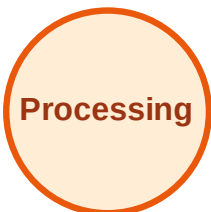
Stack (top at right): A
Visited: (none)



DFS Traversal

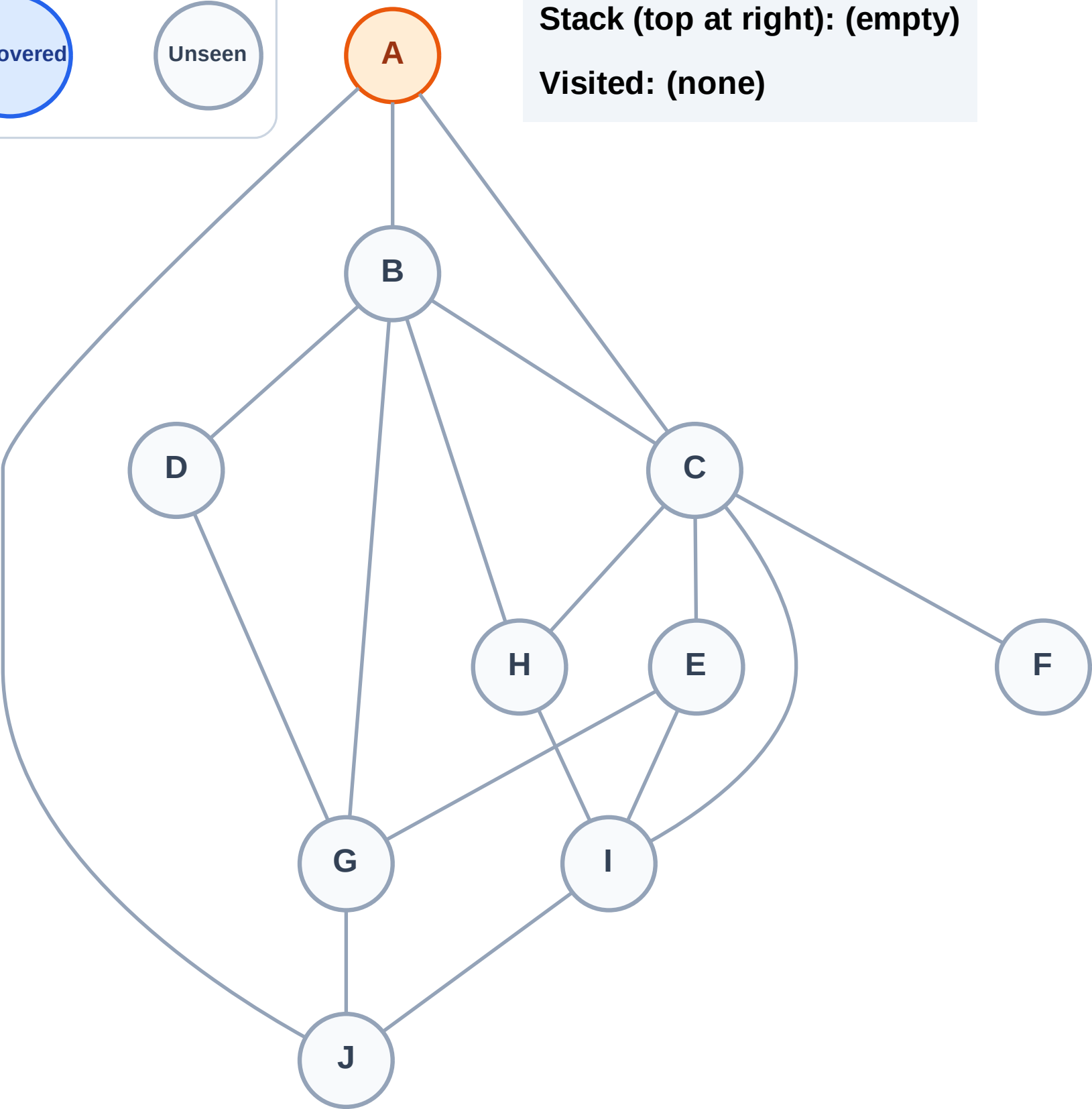
Step 2: Process node A

Legend



Stack (top at right): (empty)

Visited: (none)



DFS Traversal

Step 3: Discover J, C, B from A

Legend

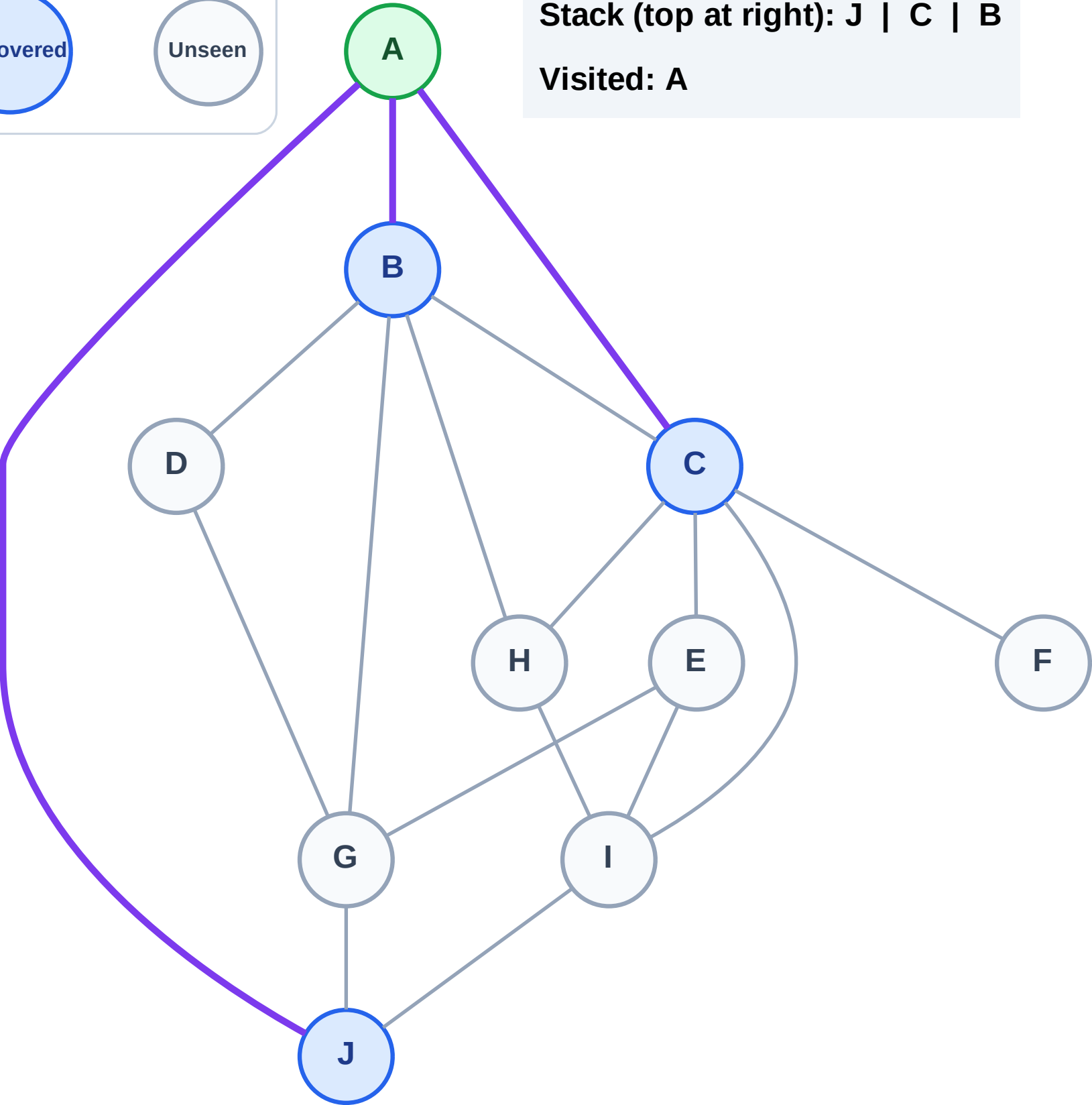
Complete

Processing

Discovered

Unseen

Stack (top at right): J | C | B
Visited: A



DFS Traversal

Step 4: Process node B

Legend

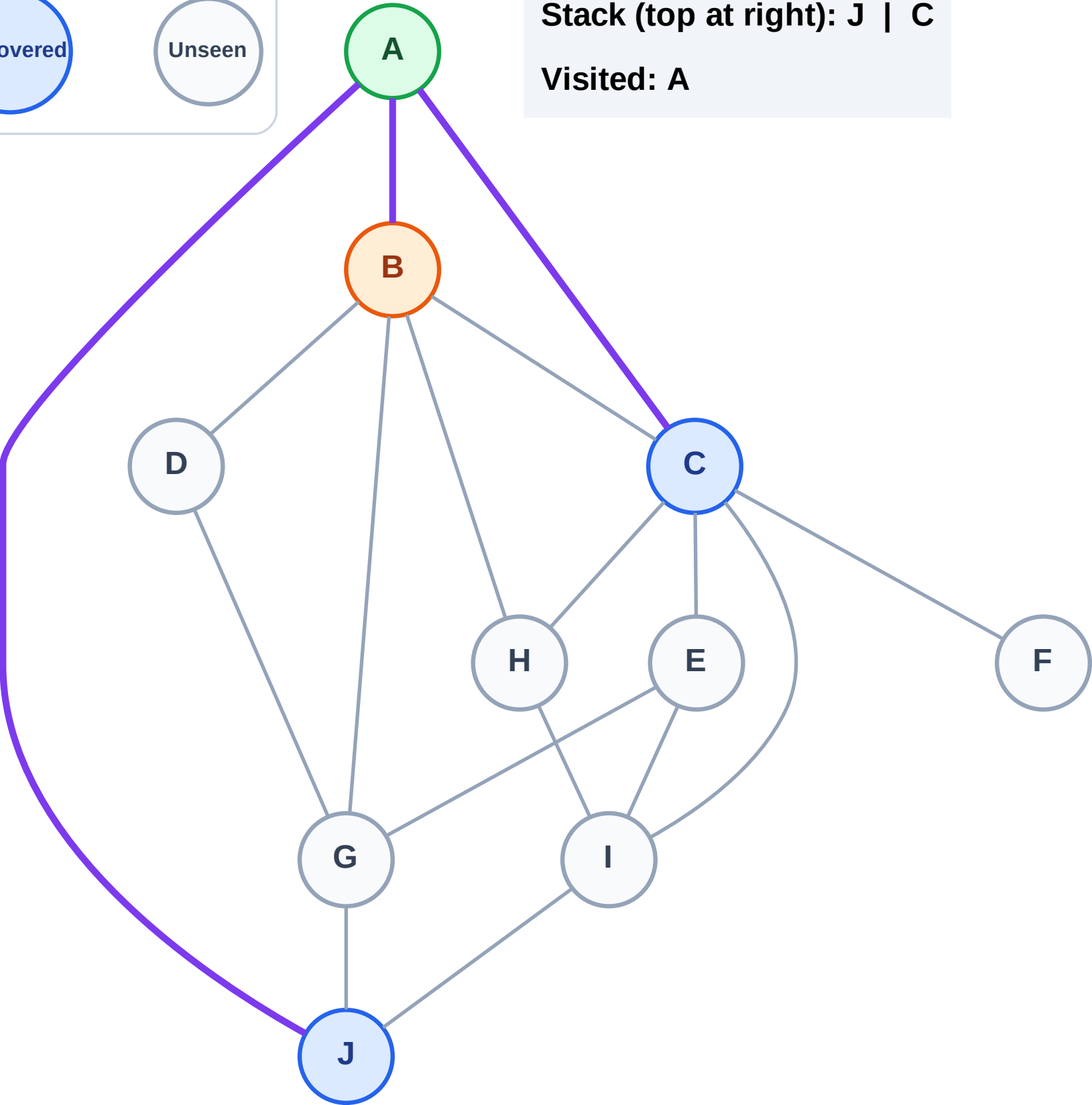
Complete

Processing

Discovered

Unseen

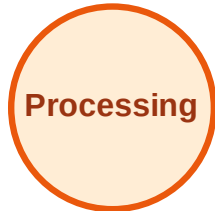
Stack (top at right): J | C
Visited: A



DFS Traversal

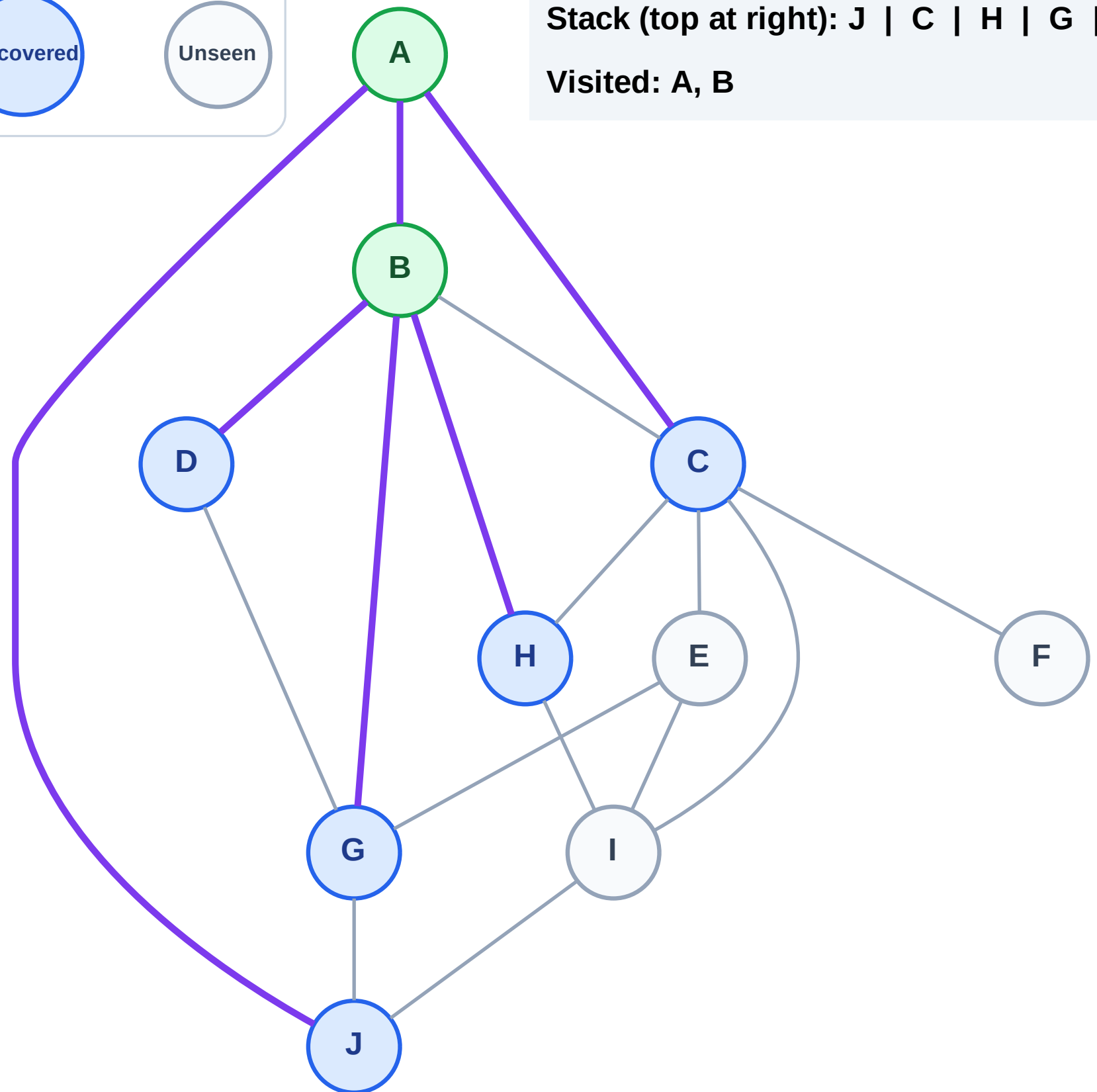
Step 5: Discover H, G, D from B

Legend



Stack (top at right): J | C | H | G | D

Visited: A, B



DFS Traversal

Step 6: Process node D

Legend

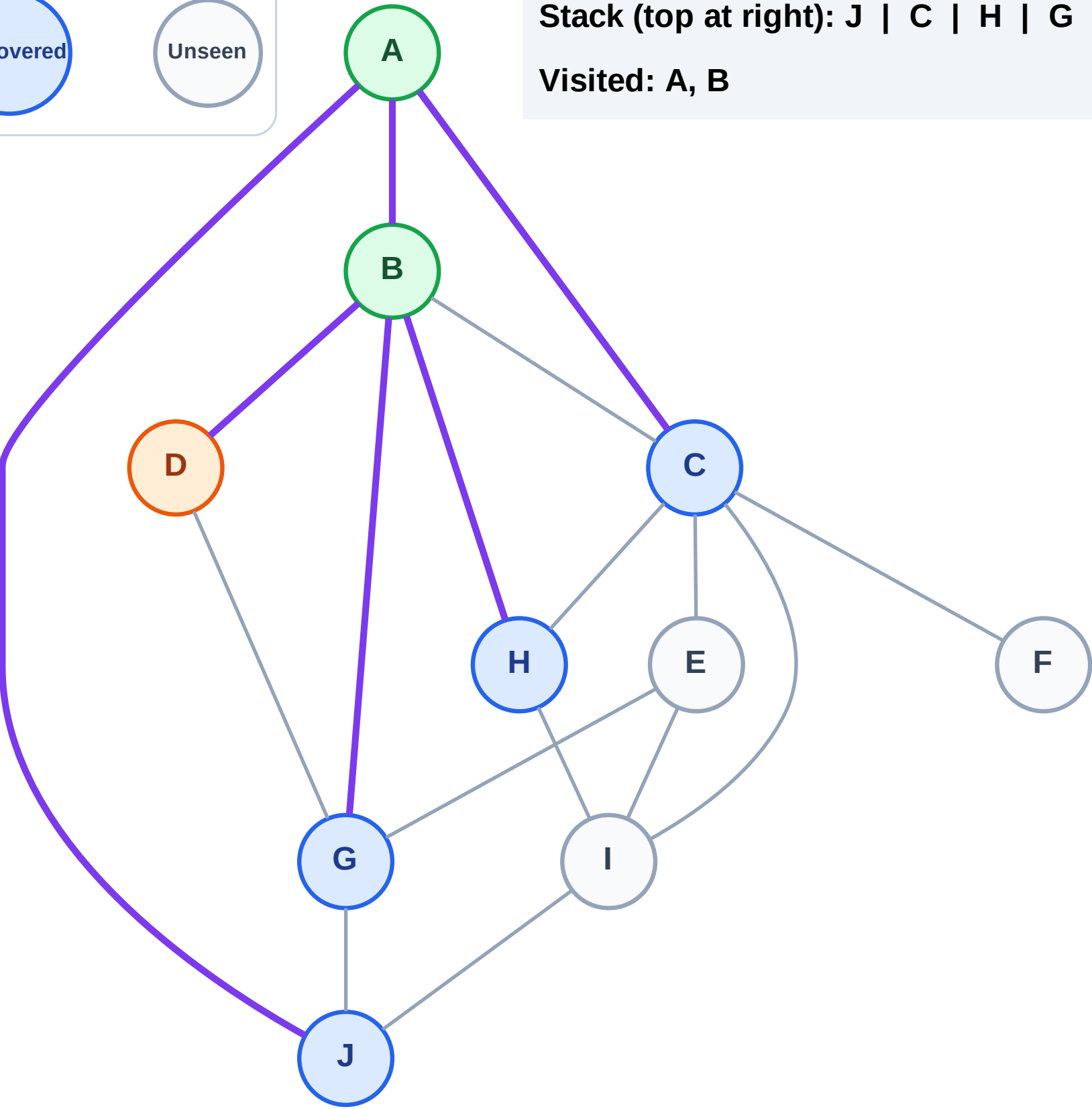
Complete

Processing

Discovered

Unseen

Stack (top at right): J | C | H | G
Visited: A, B



DFS Traversal

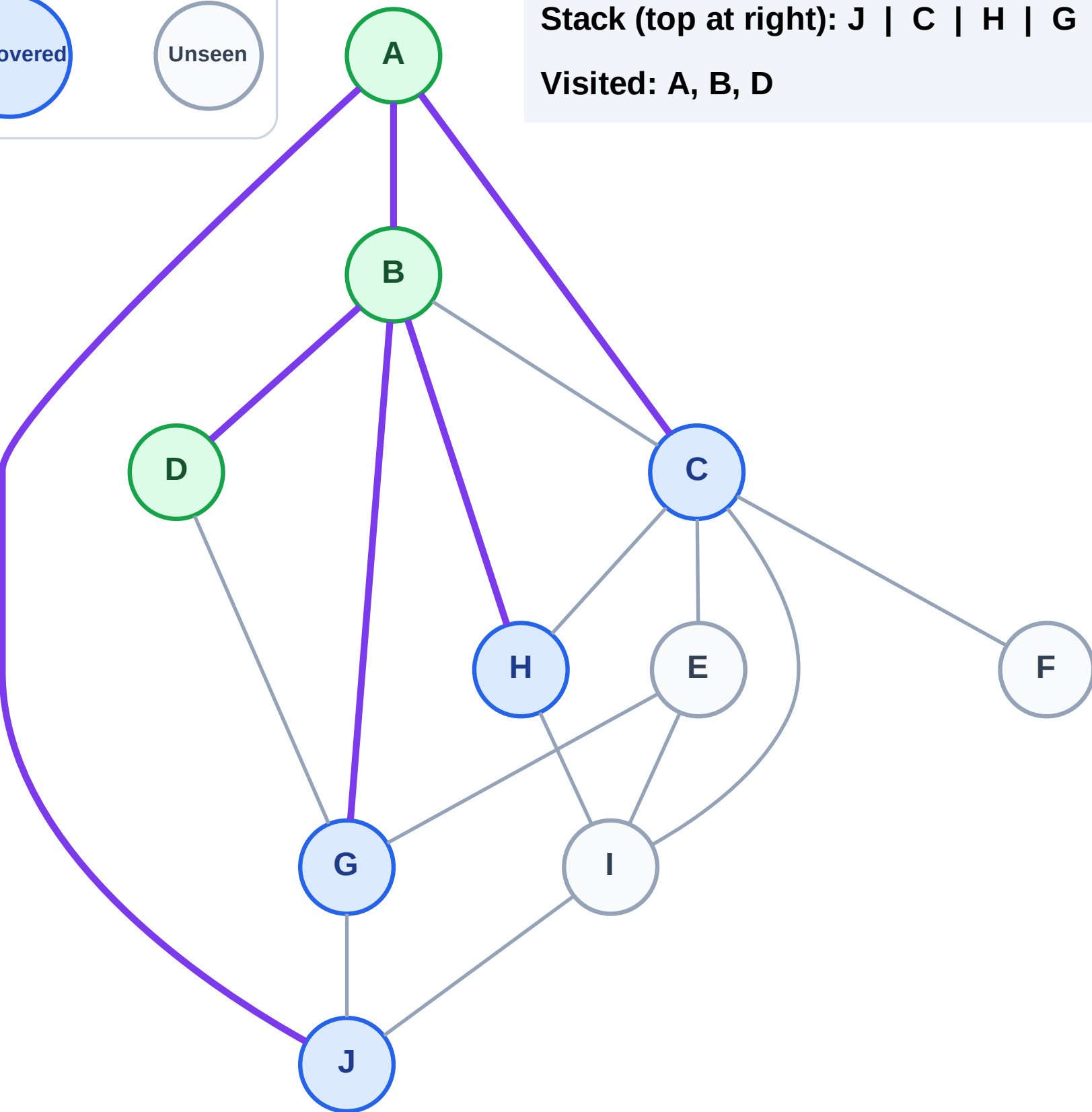
Step 7: No new neighbors from D

Legend

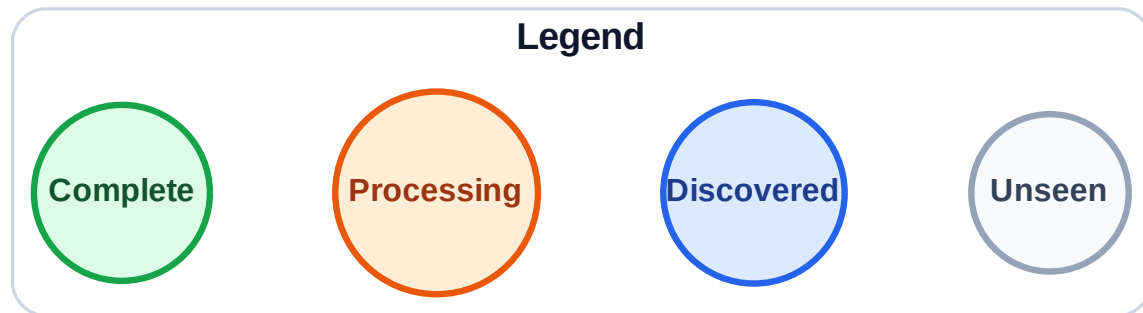


Stack (top at right): J | C | H | G

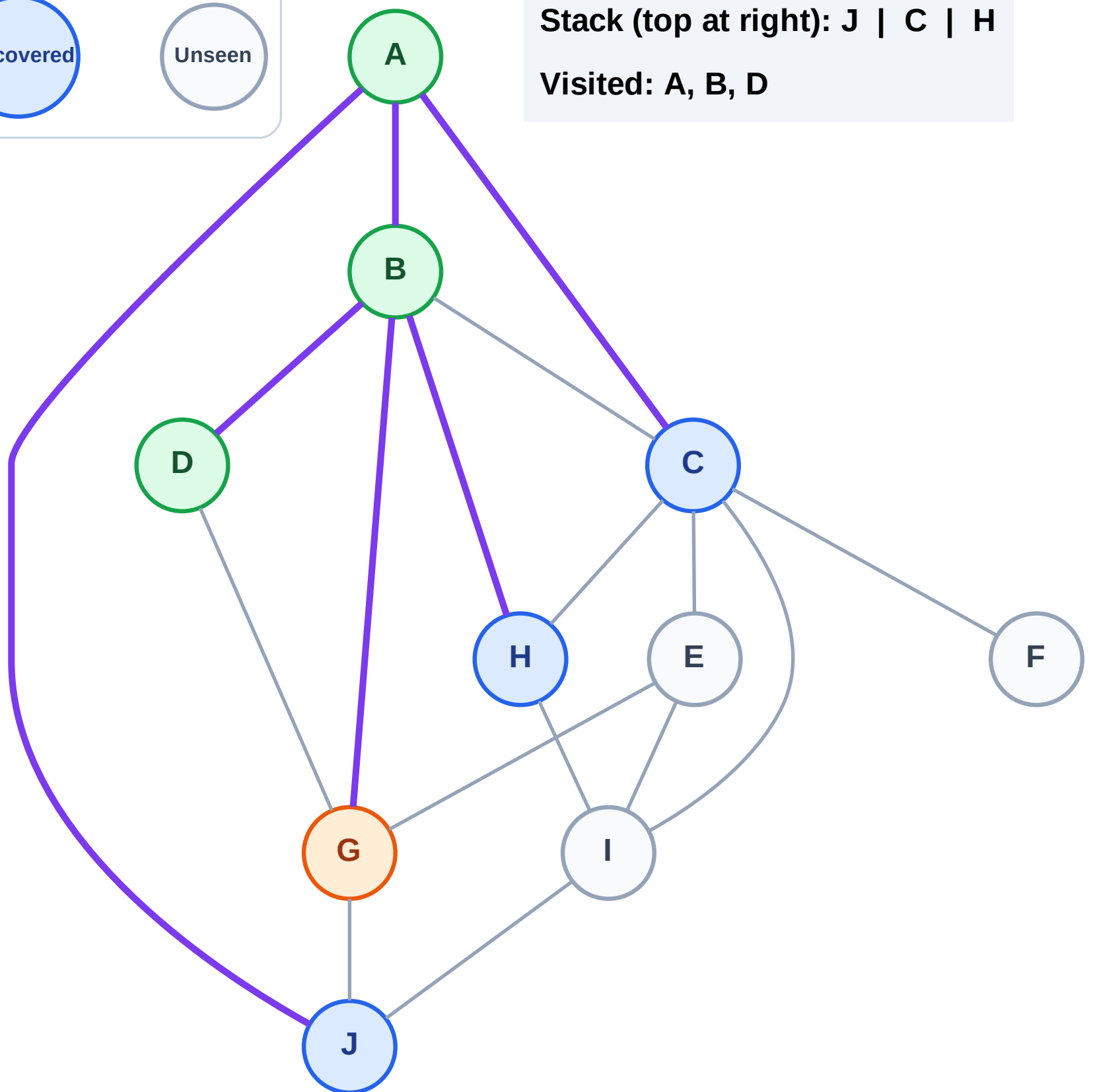
Visited: A, B, D



Step 8: Process node G



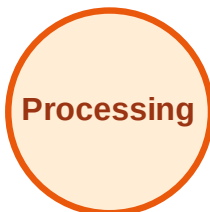
Visited: A, B, D



DFS Traversal

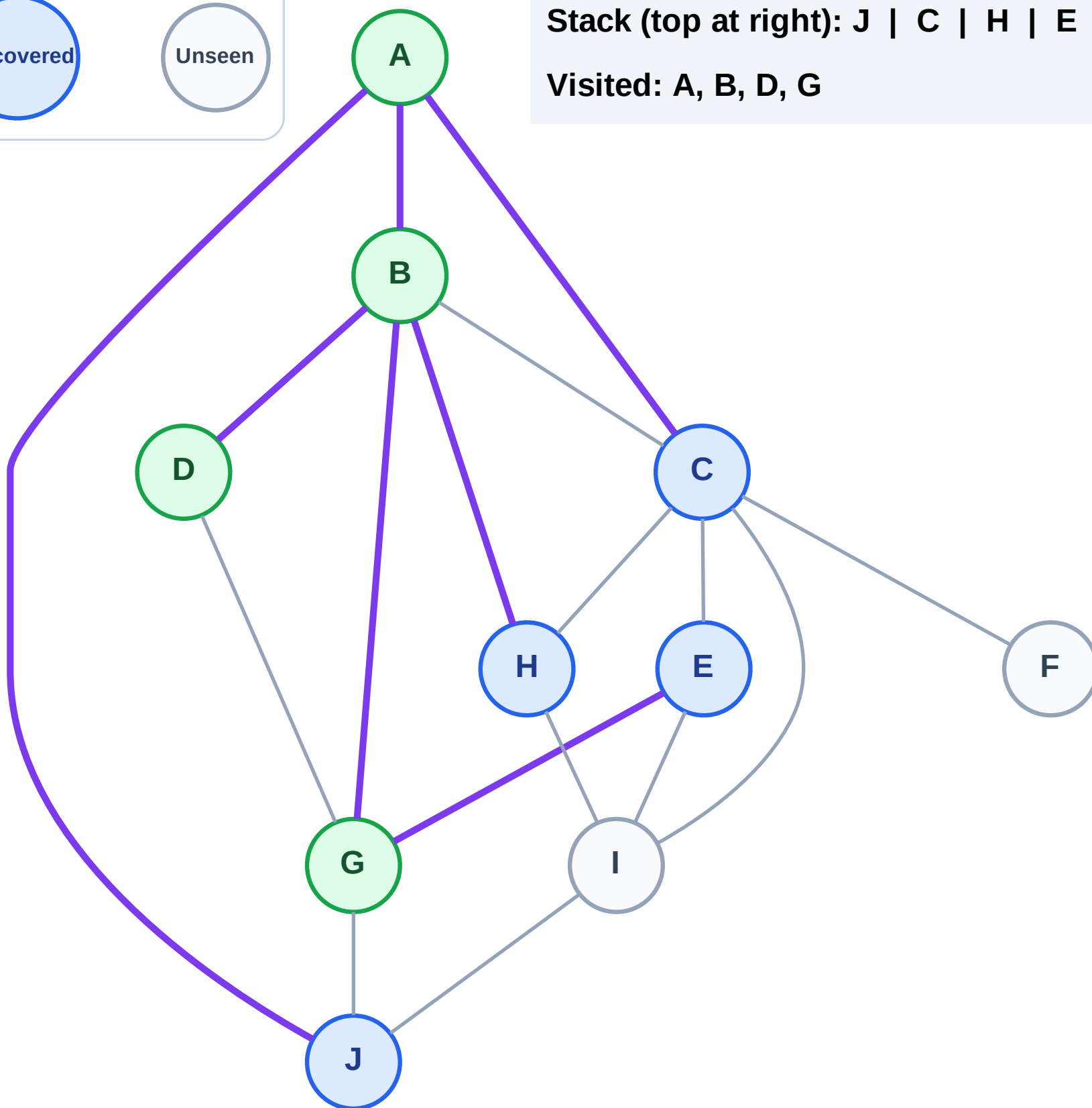
Step 9: Discover E from G

Legend



Stack (top at right): J | C | H | E

Visited: A, B, D, G



DFS Traversal

Step 10: Process node E

Legend

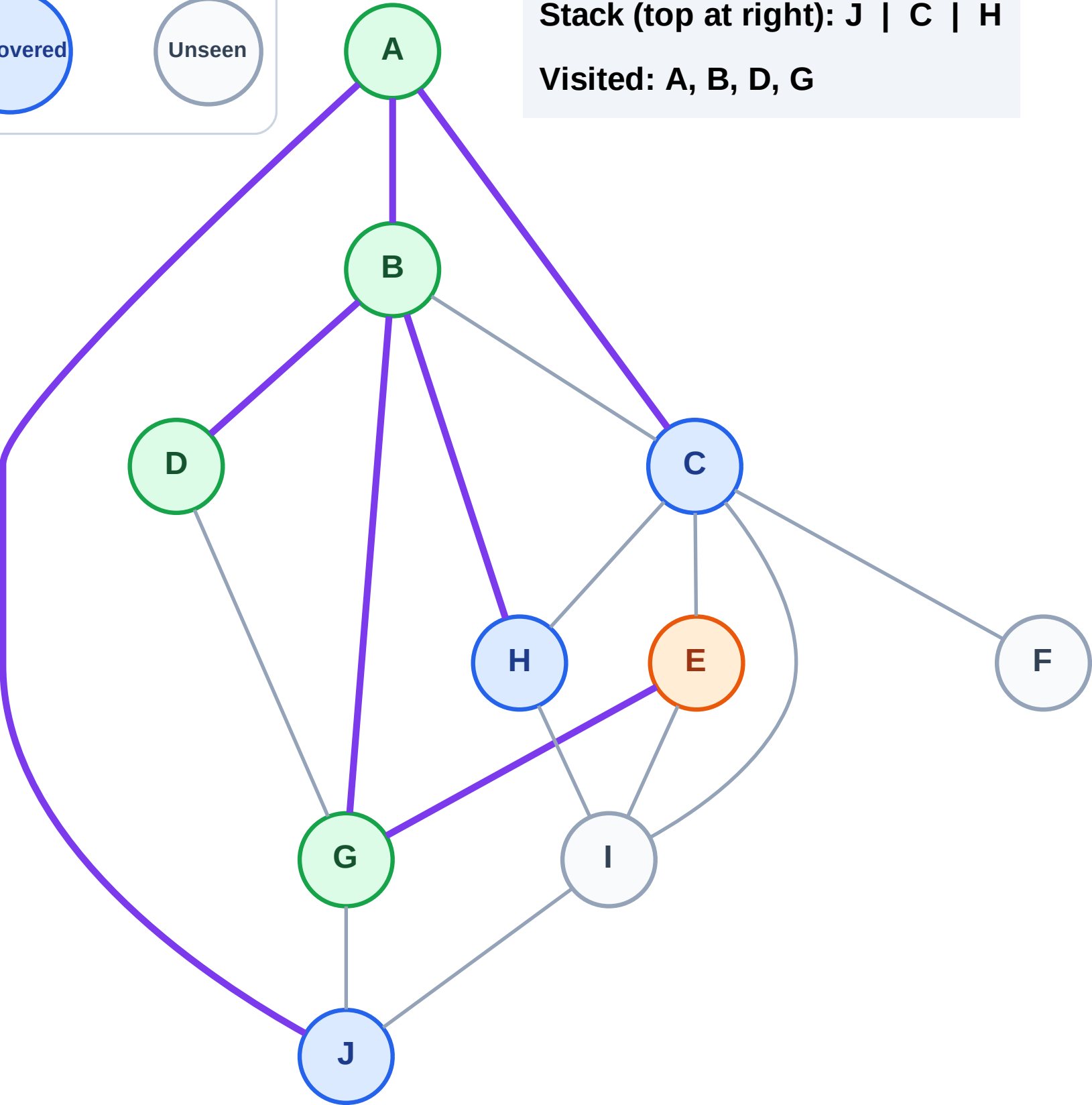
Complete

Processing

Discovered

Unseen

Stack (top at right): J | C | H
Visited: A, B, D, G



DFS Traversal

Step 11: Discover I from E

Legend

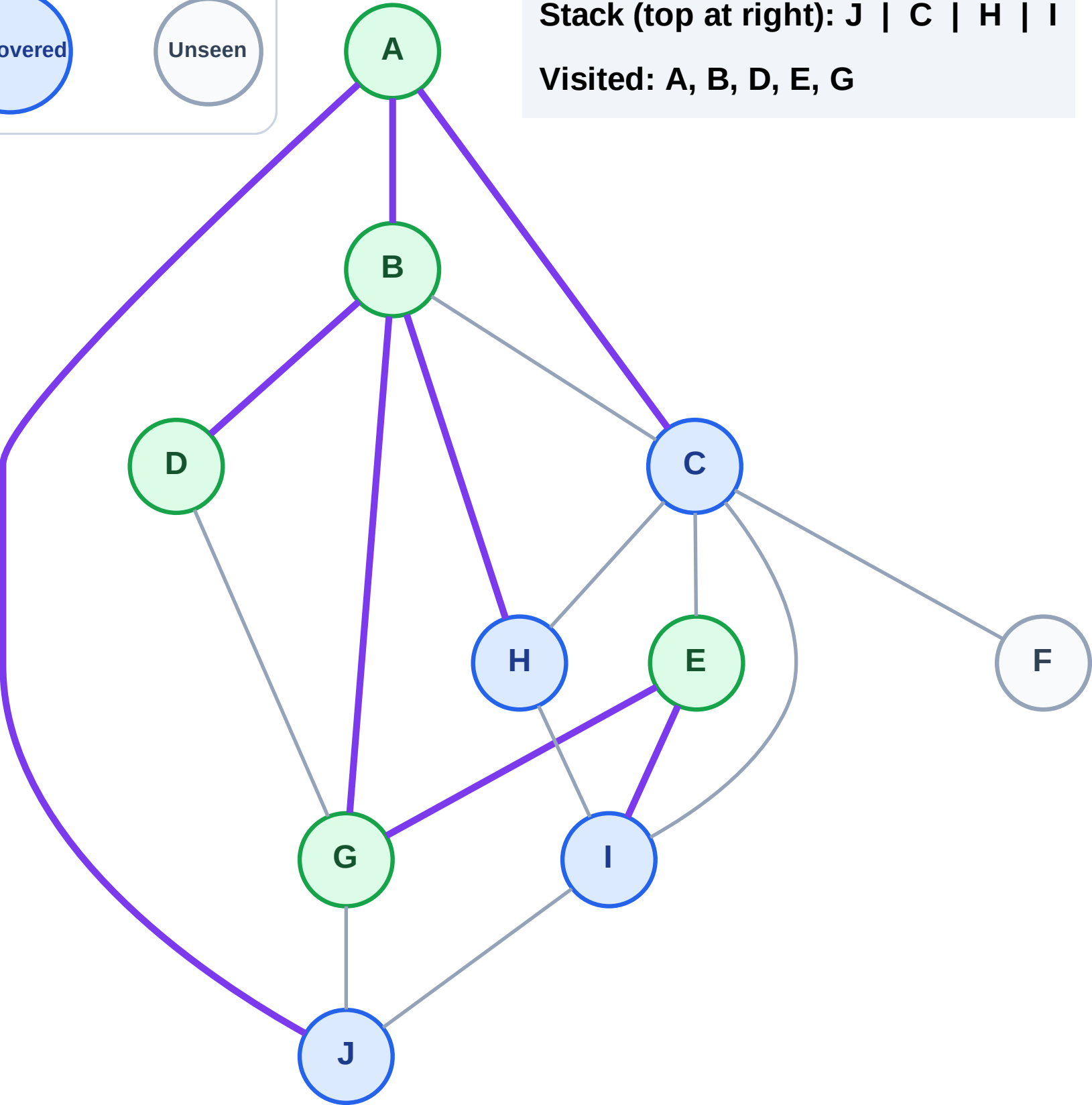
Complete

Processing

Discovered

Unseen

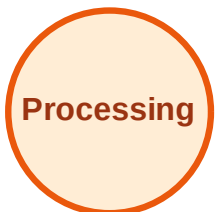
Stack (top at right): J | C | H | I
Visited: A, B, D, E, G



DFS Traversal

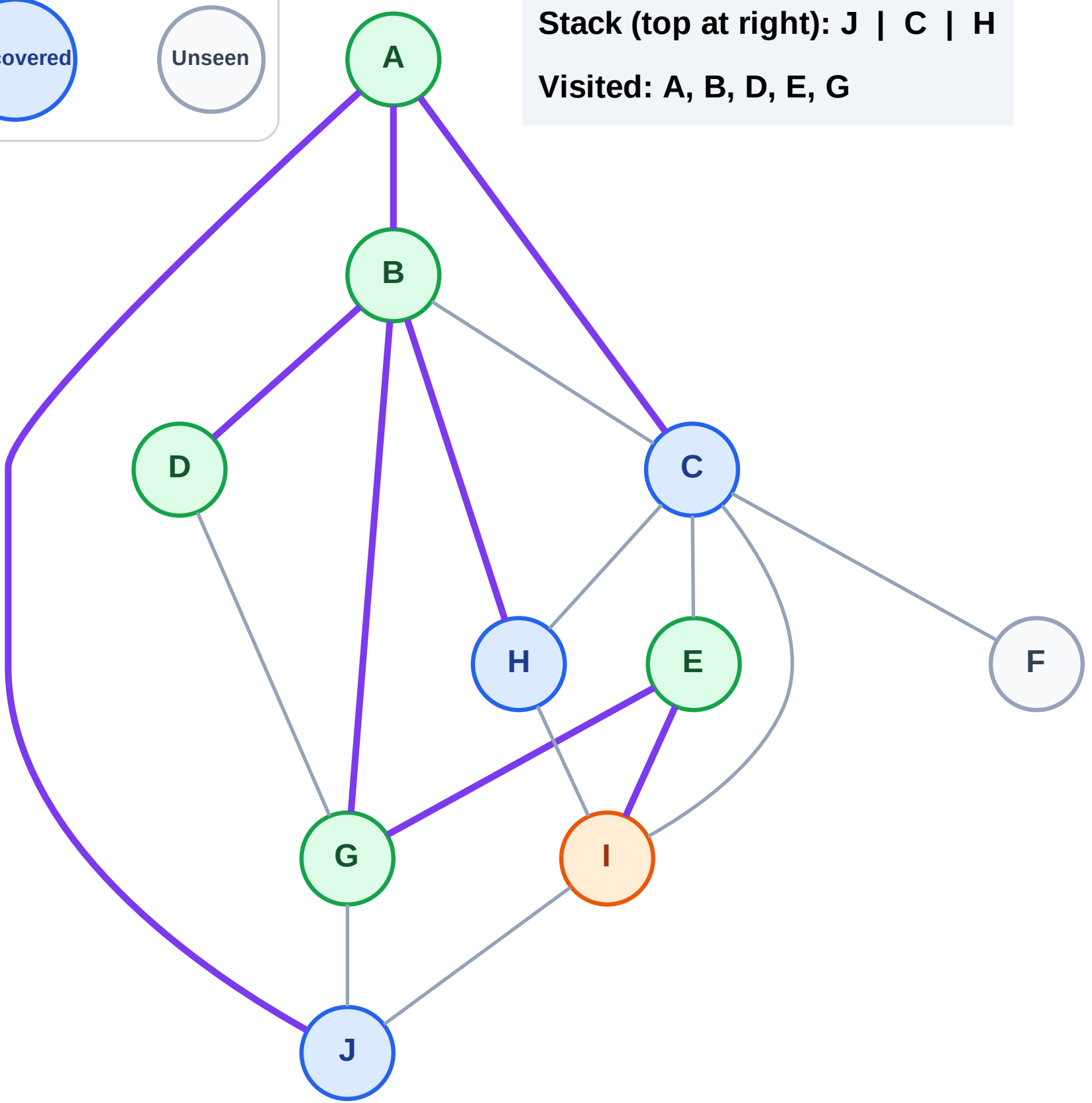
Step 12: Process node I

Legend



Stack (top at right): J | C | H

Visited: A, B, D, E, G



DFS Traversal

Step 13: No new neighbors from I

Legend

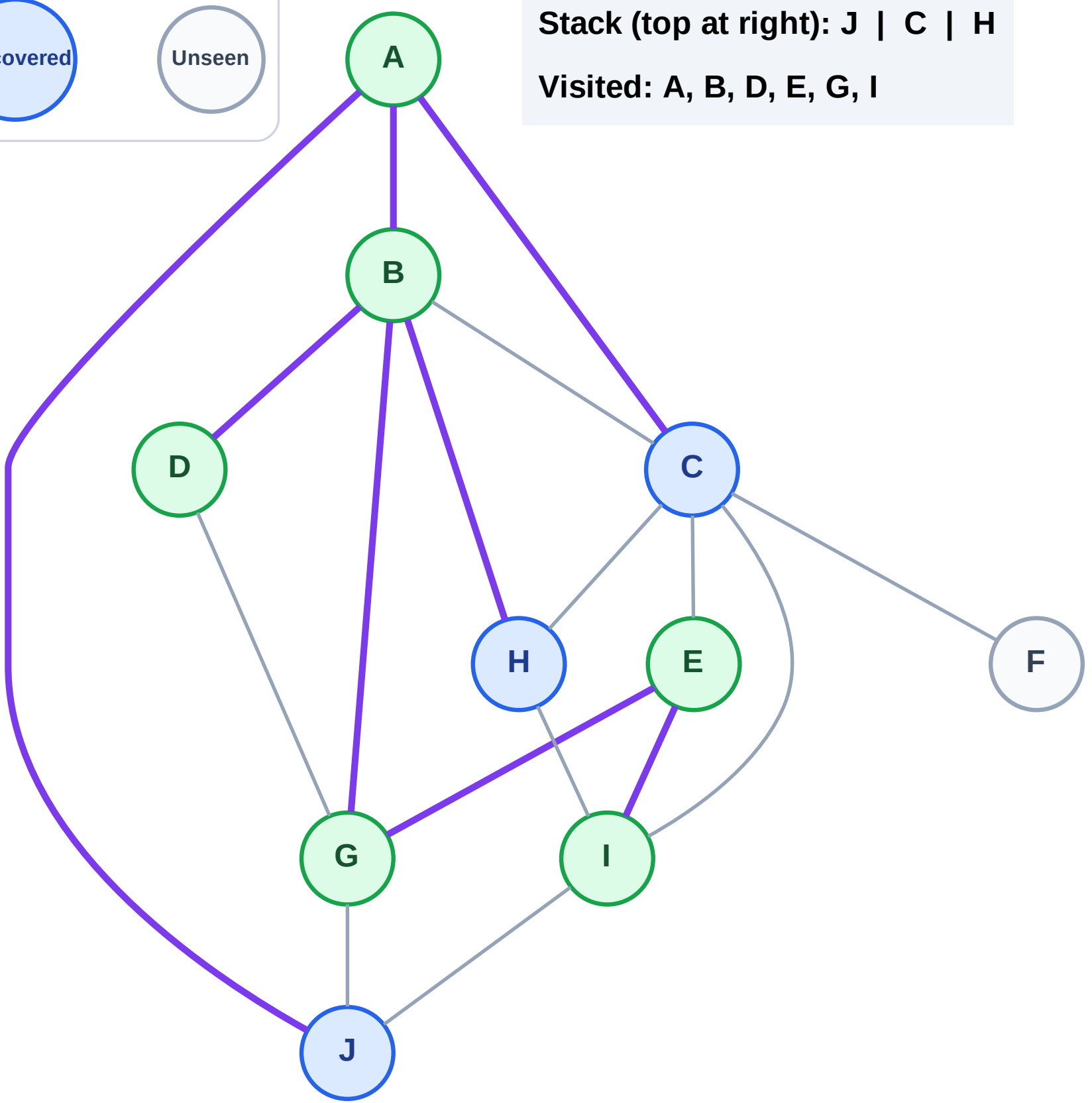
Complete

Processing

Discovered

Unseen

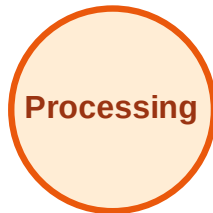
Stack (top at right): J | C | H
Visited: A, B, D, E, G, I



DFS Traversal

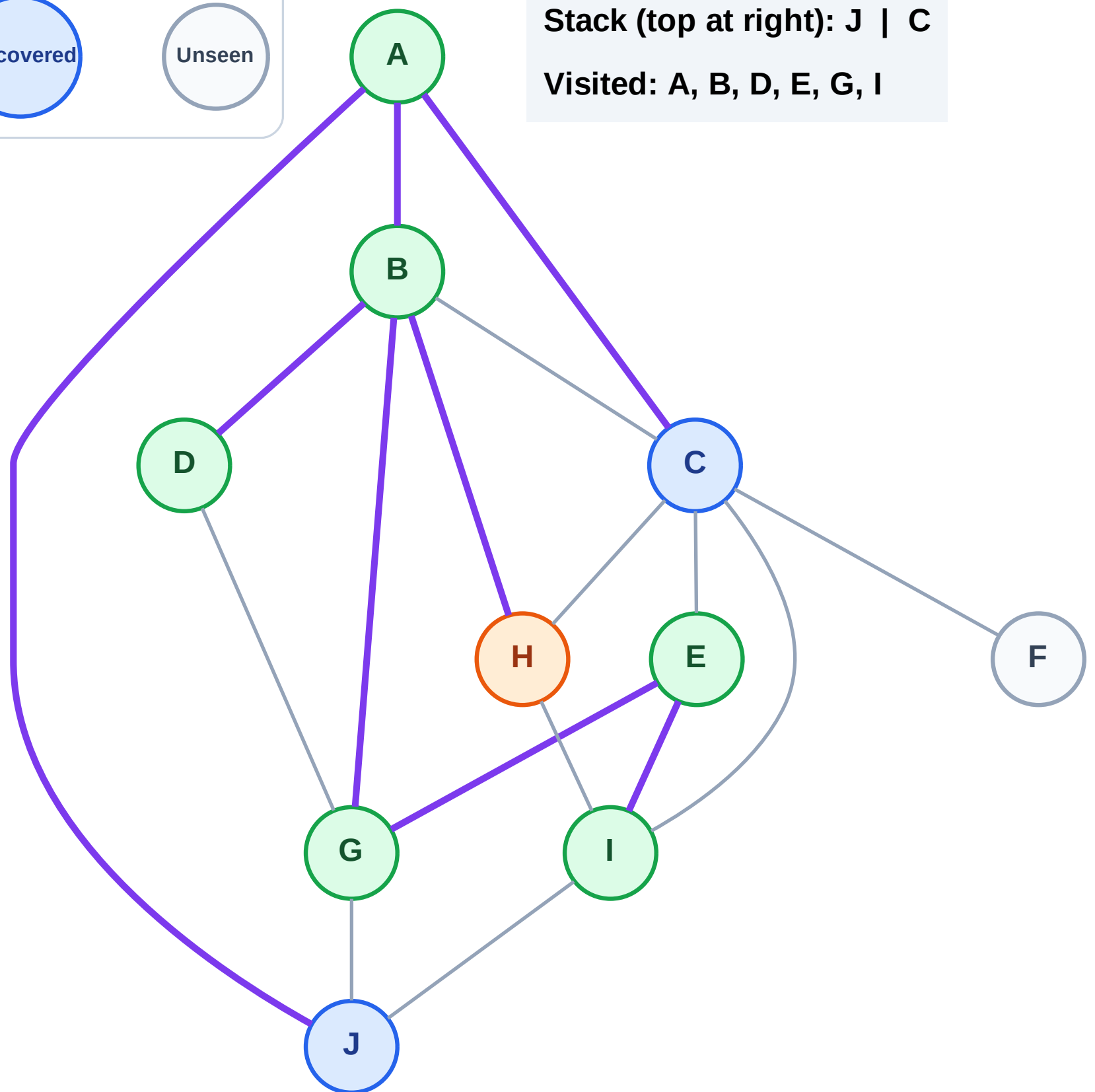
Step 14: Process node H

Legend



Stack (top at right): J | C

Visited: A, B, D, E, G, I



DFS Traversal

Step 15: No new neighbors from H

Legend

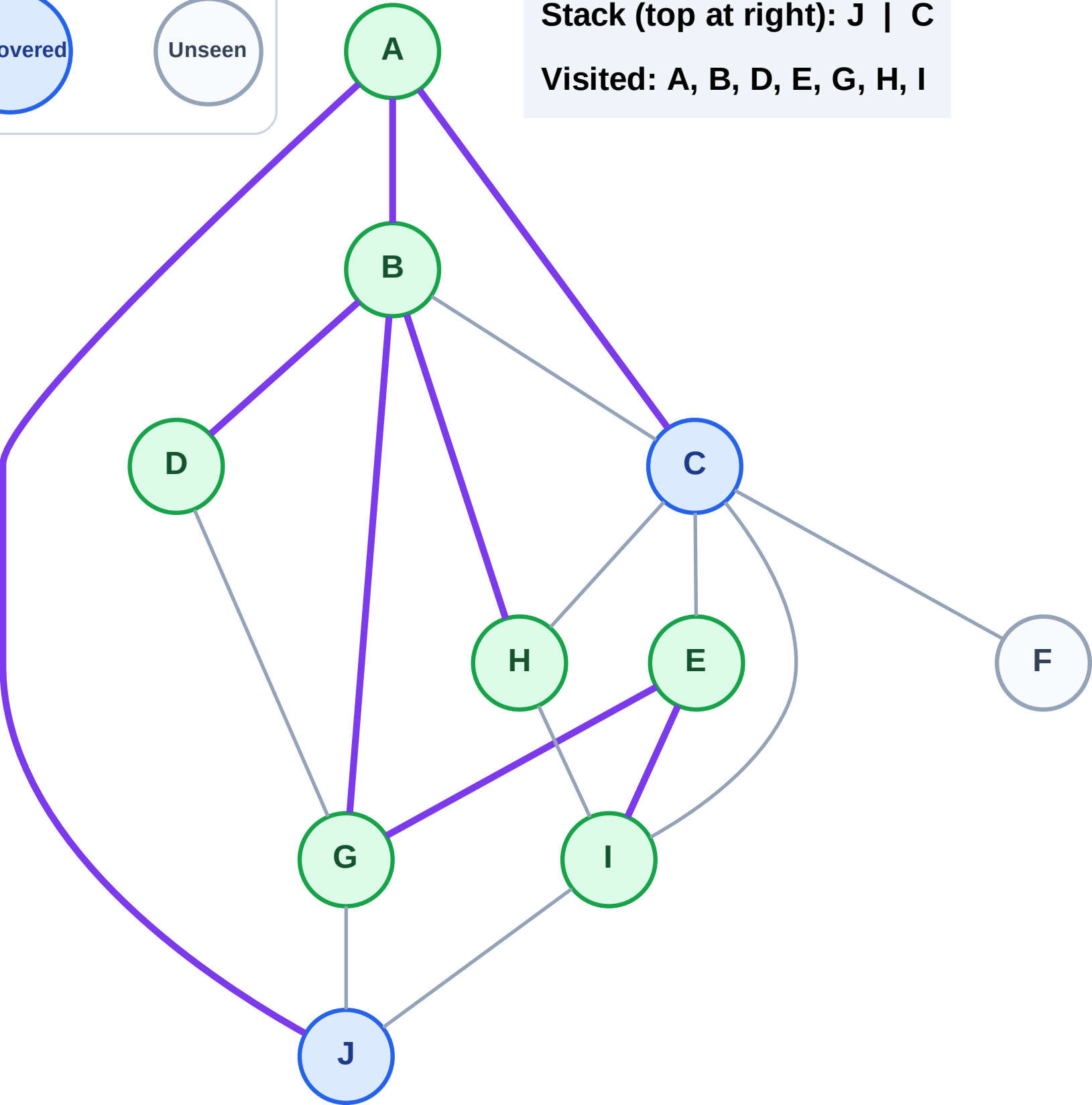
Complete

Processing

Discovered

Unseen

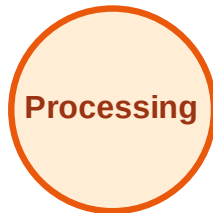
Stack (top at right): J | C
Visited: A, B, D, E, G, H, I



DFS Traversal

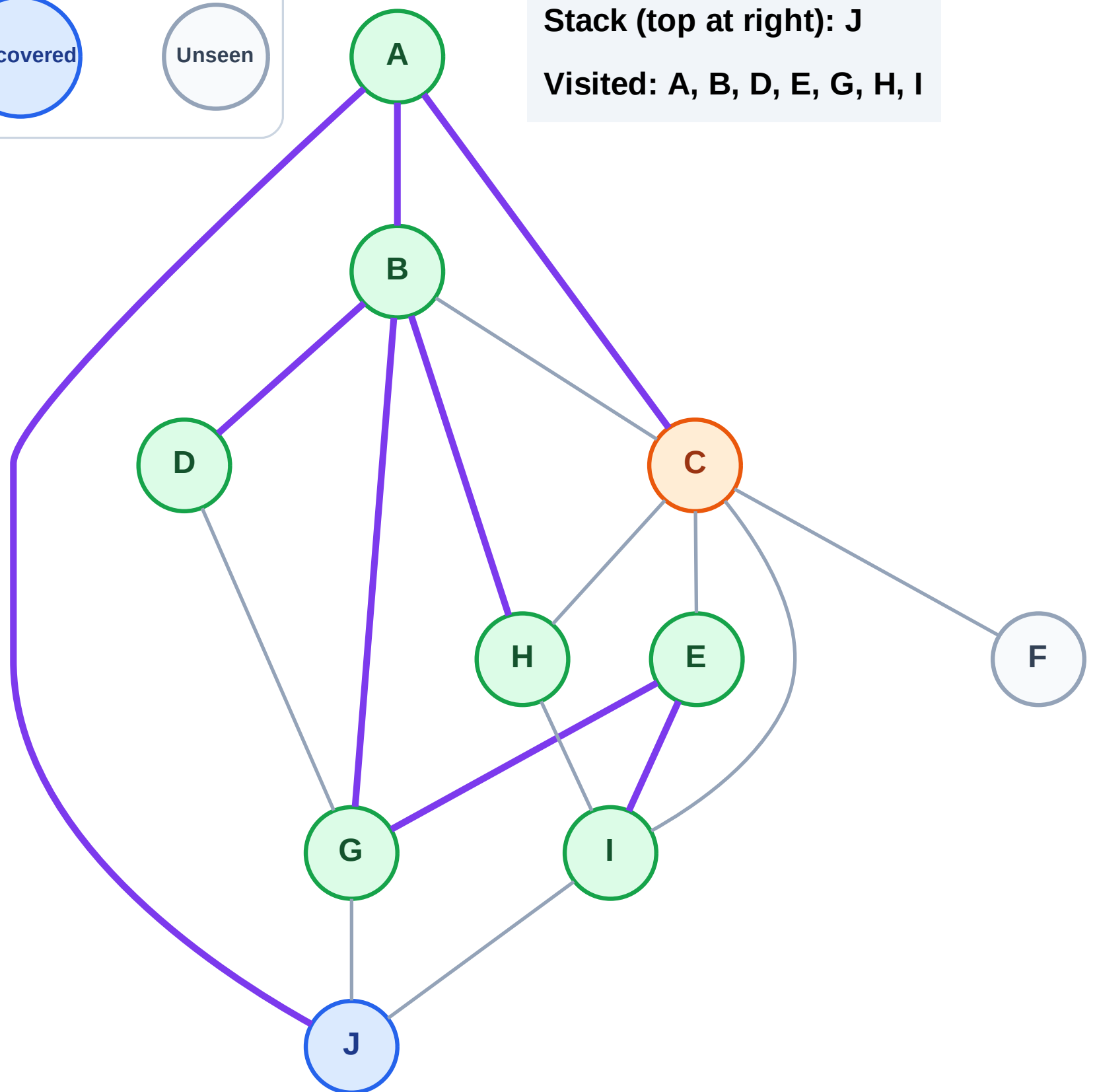
Step 16: Process node C

Legend



Stack (top at right): J

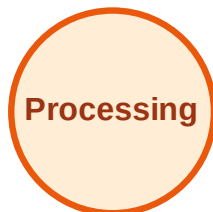
Visited: A, B, D, E, G, H, I



DFS Traversal

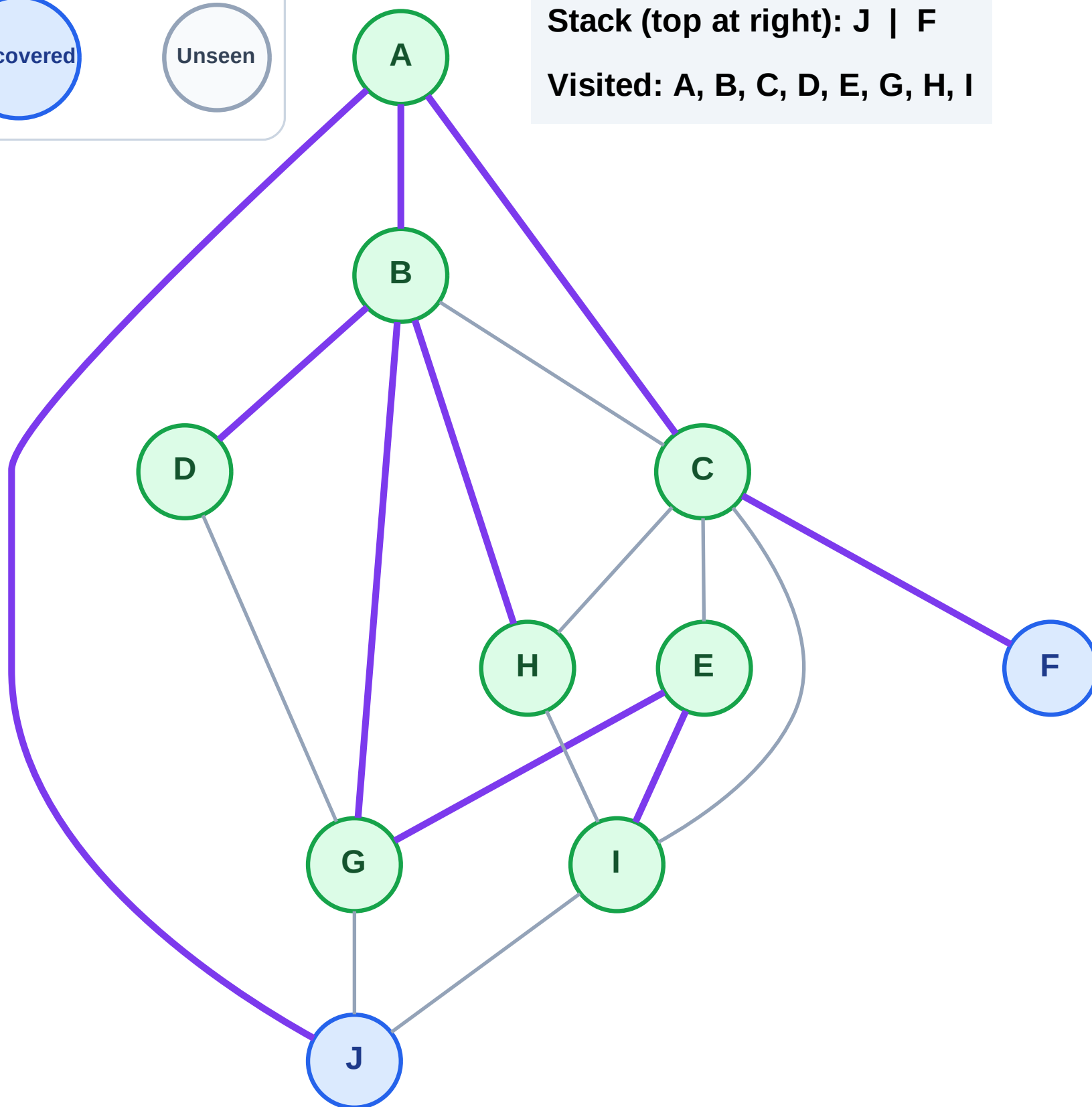
Step 17: Discover F from C

Legend



Stack (top at right): J | F

Visited: A, B, C, D, E, G, H, I



DFS Traversal

Step 18: Process node F

Legend

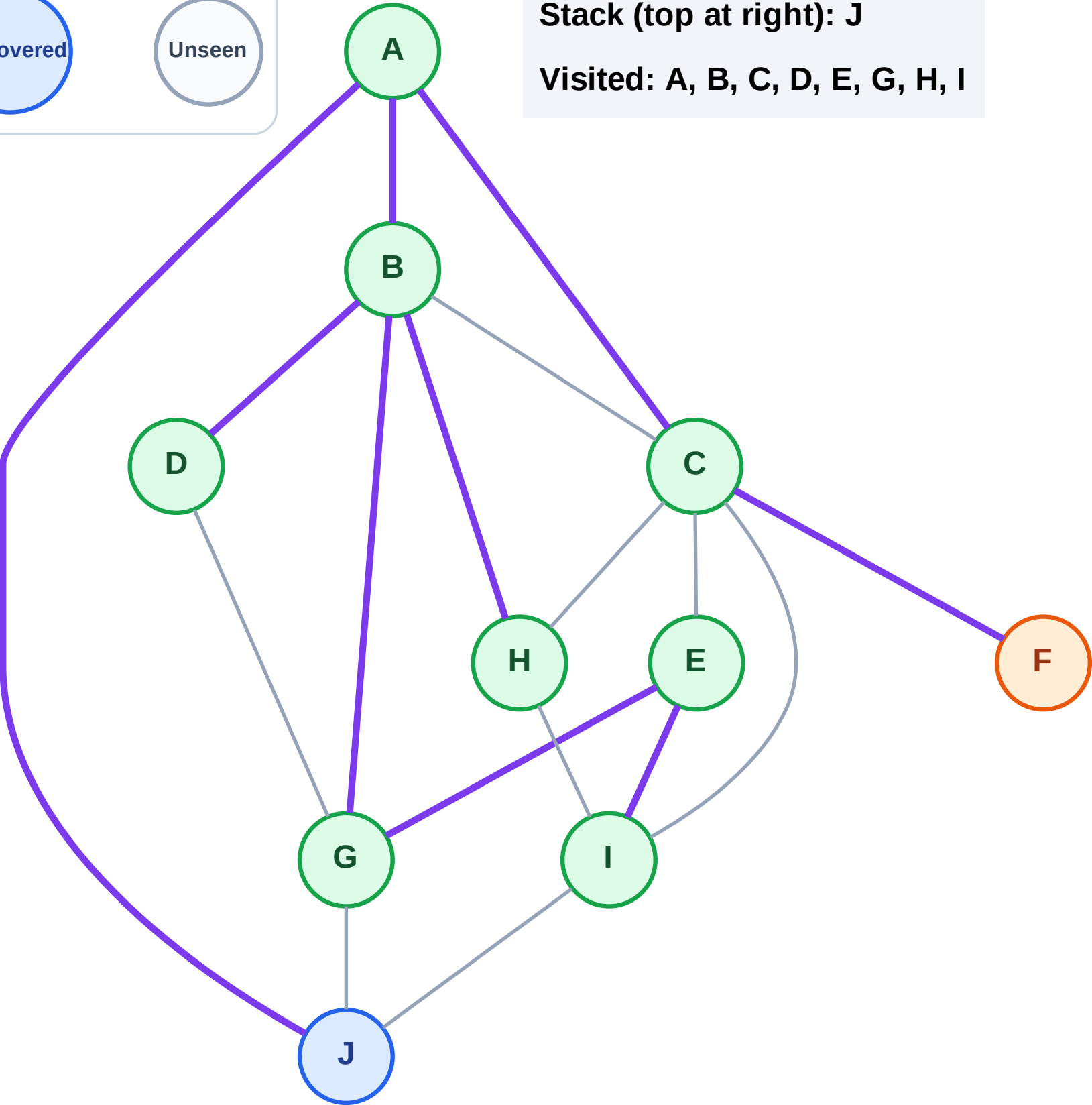
Complete

Processing

Discovered

Unseen

Stack (top at right): J
Visited: A, B, C, D, E, G, H, I



DFS Traversal

Step 19: No new neighbors from F

Legend

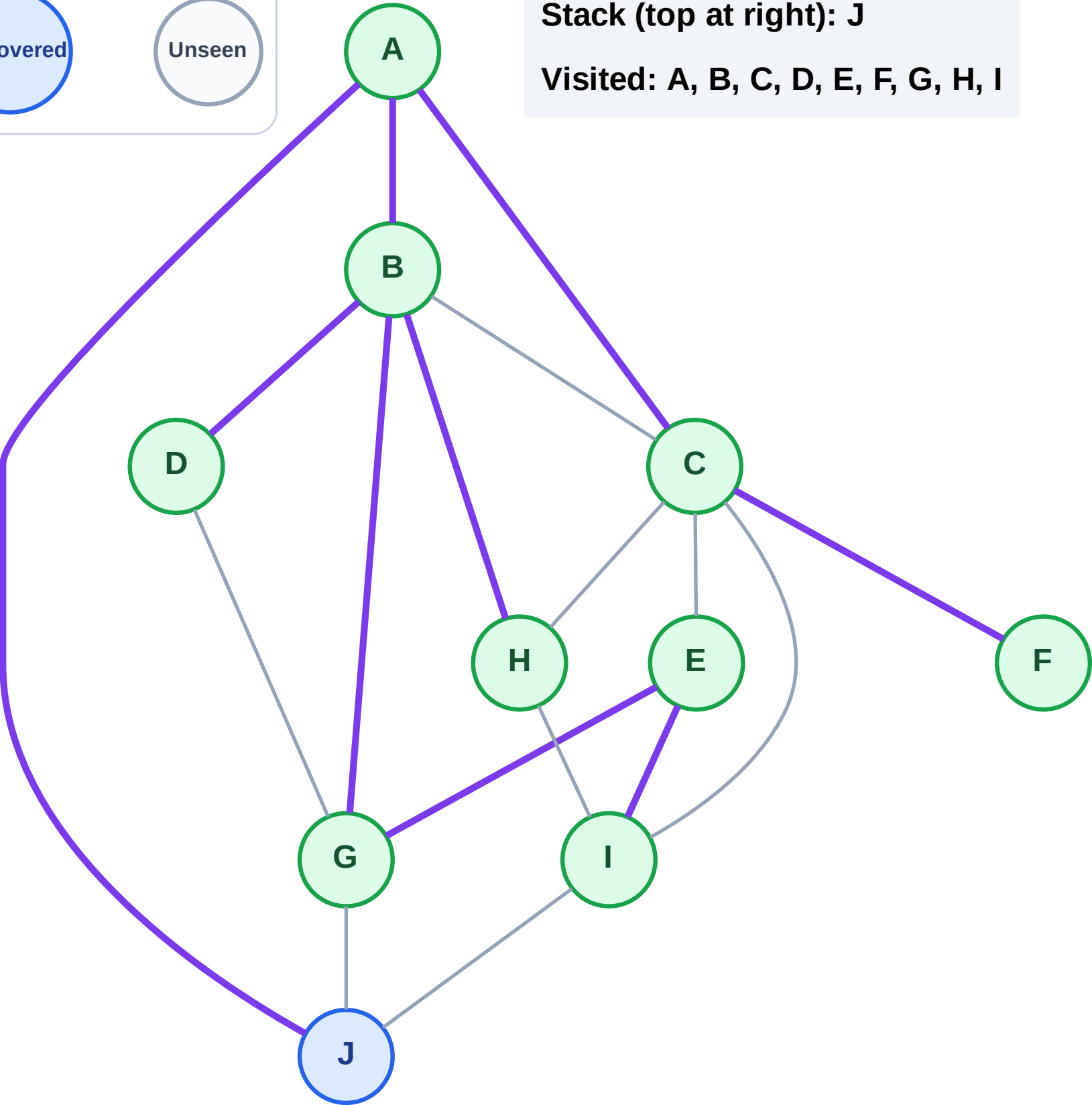
Complete

Processing

Discovered

Unseen

Stack (top at right): J
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 20: Process node J

Legend

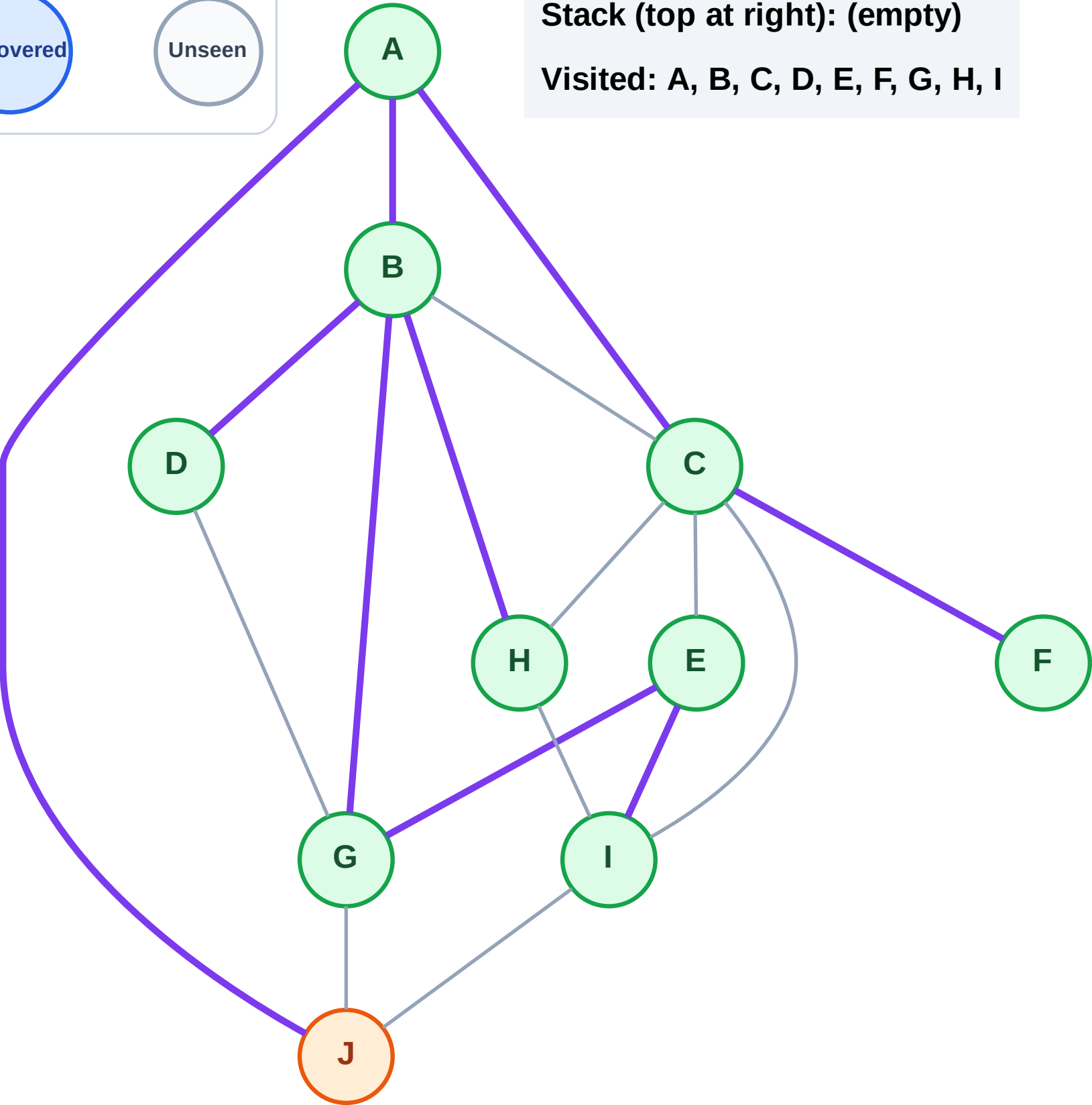
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, I, J

